Impact Evaluation of Research & Development Programs: Data and Methodological Considerations for CGIAR Research Program on Livestock

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CGIAR is a global partnership that unites organizations engaged in research for a food-secure future. The CGIAR Research Program on Livestock provides research-based solutions to help smallholder farmers, pastoralists and agro-pastoralists transition to sustainable, resilient livelihoods and to productive enterprises that will help feed future generations. It aims to increase the productivity and profitability of livestock agri-food systems in sustainable ways, making meat, milk and eggs more available and affordable across the developing world. The Program brings together five core partners: the International Livestock Research Institute (ILRI) with a mandate on livestock; the International Center for Tropical Agriculture (CIAT), which works on forages; the International Center for Research in the Dry Areas (ICARDA), which works on small ruminants and dryland systems; the Swedish University of Agricultural Sciences (SLU) with expertise particularly in animal health and genetics and the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) which connects research into development and innovation and scaling processes.

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Acronyms and Key Terms

AFAWA	Animal Feed Analysis Web Application
AMR	Anti-Microbial Resistance
ASF	African Swine Fever
BER	Basic Efficiency Resource
BPAT	Breeding Program Assessment Tool
CASH	Commercial Agriculture for Smallholder Farmers
CBBP	Community Based Breeding Programs
CBPP	Contagious Bovine Pleuro-Pneumonia
CCPP	Contagious Caprine Pleuro-Pneumonia
CEA	Cost-Effectiveness Analysis
DALY	Disability Adjusted Life Years
ECF	East Coast Fever
FFP	Food Fortification Programme
IE	Impact Evaluation
ILRI	International Livestock Research Institute
IV	Instrumental Variable
KIT	Royal Tropical Institute
Livestock CRP	The CGIAR Research Program on Livestock
LSIL	Labs for Livestock Systems
MCA	Multi-Criteria Analysis
NARS	Latin American National Agricultural Research Systems
NIRS	Near-Infrared Spectroscopy
PETS	Public Expenditure Tracking Surveys
PPR	Peste Des Petits Ruminants
PSM	Precisely Propensity Score Matching
QALY	Quality Adjusted Life Years
QEA	Qualitative Effectiveness Assessment
RCT	Randomized Control Trials
RD	Resgression Discontinuity
SDG	Sustainable Development Goals
SROI	Social Return on Investment
TOC	Theory of Change
VFM	Value for Money
WELI	Women's Empowerment in Livestock Index

Executive Summary

Background

Since the launch of the Sustainable Development Goals (SDGs) of the 2030 Agenda for Sustainable Development in 2015, traditional social and economic development agendas have been changing beyond recognition, with new partnerships and efforts focused on sustainability taking shape. This has flourished debates on the impact of development programmes and initiatives, especially because of the increasing complexity and interdependencies of development challenges as well as SDGs. A principal question for many has been, "*How do we know if we are achieving these goals*?". That is, it has been increasingly recognized that evaluation should play a crucial role to support effective and efficient implementation of research and development interventions. In this context, the SDG 16 recognizes that "*effective, accountable and inclusive institutions are essential to achieving the SDGs*". Therefore, in addressing issues of making development more sustainable, policymakers are increasingly seeking answers to the questions of what works, where, for whom, and under what circumstances in a labyrinth of new policies, program, and interventions. Impact evaluation offers evidence-based learning to governments, development agencies, and civil society on how policies and programmes delivered results and what needs to be done differently. Furthermore, it provides these organizations with better means for learning from past experience, planning and allocating resources, improving service delivery, and demonstrating results as part of accountability to key stakeholders.

In accordance with its mission, the CGIAR Research Program on Livestock[1] (Livestock CRP) aims to create a well-nourished, equitable and environmentally healthy world through livestock research for development. To this end, Livestock CRP mobilizes financial resources to fund its annual research portfolio, which spans across Africa, the Middle East, South East Asia and South and Central America, to provide research-based solutions to help smallholder farmers, pastoralists and agro-pastoralists transition to sustainable, resilient livelihoods and to productive enterprises that will help feed future generations. To design and implement a full-fledged impact evaluation of its research-based interventions, Livestock CRP commissioned this study and tasked a research team from the Swedish University of Agricultural Sciences (SLU) to ***i***) review existing methods and available approaches in the literature for assessing the impact of research investments and development interventions; ***ii***) develop a value for money (VfM) assessment framework for evaluating research activities carried out during the current cycle of Livestock CRP; ***iii***) conduct a rapid ex-post VfM assessment of the various streams of research carried under the CRP Livestock during the current cycle of the program to "*explore*" whether and the extent to which the program has had the desired outcomes and effects on the beneficiaries; and ***iv***) identify data requirements and important methodological considerations for conducting a full-fledged VfM assessment of Livestock CRP.

Methodology and limitations of the evaluation

To provide the CRP Livestock with timely answer in less than three months, the study team addressed the abovementioned 4 objectives by relying mainly on the literature and the annual reports of CRP Livestock from 2016 to 2020. Specifically, in relation to objectives ***i*** and ***ii***, the study team adopted a rapid scoping review methodology, based on Arksey and O'Malley (2005), to review existing methods and available approaches in the literature relevant to ex-post impact evaluations of research investments and development interventions. Scoping reviews describe existing literature and other

[1] Read more about CRP livestock here: https://livestock.cgiar.org/about/why-livestock

sources of information commonly include findings from a range of different study designs and methods. Munn et al. (2018) illustrate that a (rapid) scoping review can be a particularly useful approach when the information on a topic has not been comprehensively reviewed or is complex and diverse. Thus, a rapid scoping review framework can help achieve several objectives, including identifying types of existing evidence in a given field, clarifying key concepts or definitions in the literature, and surveying how research is conducted on a certain topic. The rapid scoping review was carried out by searching Google Scholar, Scopus, organizational websites and internet searches. Eligible publications included peer-reviewed articles, reviews, reports and policy documents published in English from January 2000 to August 2021. It should be noted that rapid and less-formal methodology adopted in scoping reviews is a clear limitation and it is possible that key elements of integrated care or important implementation issues were not captured. Therefore, the list of impact evaluation tools, methods, and approaches overviewed in this report is not comprehensive, nor is it intended to be. Some of them are complementary and some are substitutes. The choice of which method or approach is appropriate for a given context would depend on a range of considerations including the use for which the impact evaluation is intended, the stakeholders interested in the findings, as well as the scope, the cost and the time by which the results are needed, etc.

With regard to objectives ***iii*** and ***iv***, the essential methodological tool was "content analysis" of the annual reports of CRP Livestock between 2016 and 2020, which was applied basically upon a qualitative and deductive approach. Typically, content analysis, as a research method, can be utilized either qualitatively or quantitatively for an objective, systematic, and quantitative description of manifest content of documents and communications (Klenke, 2008).

The main limitations of this study were the following. First, the evaluation timeframe of the study that reduced the depth and breadth of analysis. That is, the study started in August 2021 and most data collection and analyses took place in September and October. Emerging findings and recommendations were discussed with CRP Livestock. This is the third draft of the study report, which incorporates comments from CRP Livestock and its team. Second, it was not possible for the study team to access detailed information and disaggregated data on the beneficiaries of the program activities, which influenced the methodology selection and application.

Main findings and conclusions

Overall, we found that CRP Livestock has provided research-based solutions to smallholder livestock keepers in the target regions and countries, however; the magnitude of impact of the CRP livestock varies from flagship to flagship. Generally, the CRP livestock contributed to improvement in income, food and nutritional security, poverty alleviation, capacity building and improvement in livelihoods of beneficiary households and individuals including women, children and youth in the priority countries. The CRP livestock contributed to improvement in sustainable management of the environment, biodiversity and climate change adaptation in priority communities in the priority countries. Furthermore the CRP livestock contributed to capacity building, educational development and innovation of academic and research institution in partner and priority countries. We found that the direct impacts of the CRP livestock differ across research streams. it is important to highlight that impact evaluation of the CRP Livestock, like any evaluation, will largely be most reliable and valid when it uses a mixed methods approach where both qualitative and quantitative data are used to establish evidence for impact

Regarding relevance and coverage of CRP Livestock activities, we found that the design of the program is broadly relevant to the mandate of CGIAR and to the needs of the beneficiaries in the target countries. Regarding effectiveness, we found that CRP Livestock has generally made good progress against planned outputs and productivity has been high. This has included inter alia some innovative and influential work, for example on methods for varietal adoption studies. CRP livestock developed and disseminated effective and locally adapted animal genetics, crop and forage varieties, feed products, vaccines, business models, tools and technologies in the priority countries. We believe that for the next phases of the program, more thought should be given to the theory of change, and the institutional learning and communications investments needed to reach the wider outcomes desired.

Regarding efficiency, we found that the CRP livestock attained relevant efficiency gains. The first relates to the joint development of greenhouse gas assessment methodology and capacity by both CCAFS and the Livestock CRP, reducing duplication of effort and cost. Second, investment in IT solutions for information collection, management and application for detection and response to implementation problems in large scale projects is reducing replicated efforts and improving effectiveness. Efficiency gains were observed for an average animal in each system adopting different CRP technologies at scale. We believe that for the next phases of the program, more thought should be given to quantifying efficiency gains observed across different streams. In this way, the overall efficiency gain for the CRP livestock can be aggregated.

Regarding impact, it was found that the short-medium and long term impacts and its magnitude varies across flagships. We found the long term impacts of the Animal Genetics flagship include (a) contribution to poverty reduction, (b) contribution to hunger and food insecurity reduction, (c) income, (d) education, knowledge and innovation, (e) capacity development. The Livestock Health flagship in the long term contributed to (a) improved health of both human and animals (b) food/nutritional security (c) increase income of poor livestock keepers including women and youth (d) improved livelihoods livestock keeping households and communities (e) education/capacity development. The long term impacts of Feeds and Forages include (a) increase in income of poor livestock keepers (b) improved livelihood of livestock keeping households and communities (c) enhanced nutritional and food security (d) improved environmental health and biodiversity (e) increased knowledge in feed & forage science. In the long term, the Livestock and Environment flagship contributed to (a) climate adaption and mitigation (b) improved environmental health and biodiversity (c) improved natural resource conservation (d) capacity development. The Livestock Livelihoods & Agri-Food Systems in the long term contributed to (a) Women and youth empowerment (b) Nutrition & food security (c) Poverty reduction (d) Increased Income. In the next phase of program, we believe that quantification of the impact dimension of the CRP livestock program using the value for money approach will require qualitative and quantitative data on outcome indicators across streams.

In terms of sustainability, the CRP Livestock has incorporated strategies to ensure sustainability of CRP livestock streams' impacts at the end of the project. Strategies to sustain the CRP livestock include collaboration with national and international development partners, government agencies and extension systems, including technology developers. This collaboration ensures the continuation of the streams activities and its impacts beyond the end of the CRP livestock program period. Another sustainability strategy is the incorporation of CRP livestock streams' technologies, tools and concepts into teaching materials and curriculum of universities and research institutions in the priority countries. Students and researchers are the future farm advisors, policymakers and industry representatives, making it important to communicate and transfer the knowledge to future generation.

***Limitation and considerations**:* Our impact assessment of CRP livestock has some limitations. To begin with, the short timeframe did not allow a fully harmonized estimation of causality parameter estimates in terms of magnitude of impacts attributed to each research stream. Secondly, there is insufficient data to allow for quantifying the impact of the different CRP livestock streams. In addition, there is no end-line data on the outcome indicators for the different CRP livestock streams. The broad nature of the CRP livestock makes it difficult to quantify the actual impact from each product line's activity in priority countries. CRP currently does not have data on number of beneficiaries of each research stream and product lines and as such it is difficult to quantify the magnitude of impacts from the CRP streams.

With regard to considerations for future VfM assessments of CRP Livestock, it is important to highlight that impact evaluation of the CRP Livestock, like any evaluation, will largely be most reliable and valid when it uses a mixed methods approach where both qualitative and quantitative data are used to establish evidence for impact. Thus, quantification of the impact dimension of the CRP livestock program using the value for money approach will require additional qualitative and quantitative data aggregated across streams.

A livestock keeper in Ethiopia holds a kid goat. Photo ILRI/Apollo Habtamu

1. Review of some tools, methods and approaches of impact evaluation of research & development interventions

Impact evaluation (IE) differs from the traditional process evaluations that only focus on describing how a project was implemented. Instead, IE involves the application of empirical studies to quantify the causal link between interventions and outcomes of interest. Impacts can be measured at any level and, contrary to general perception, do not need to concern only long-term goals or "impacts" in the jargon of logical frameworks (White and Raitzer, 2017; Rogers et al., 2015). There is no single approach that can be used for all IEs. The most common approaches can be roughly categorized into four types (see Stern 2015). It is important to note, however, that there are often large areas of overlap between these four types. In addition, many IEs include more than one of the types. IE uses a range of different approaches and can be undertaken at any stage of a program's lifecycle. In this context, two common terms are used to describe evaluations: formative and summative. Formative evaluations are typically conducted during the program period to improve or assess the program delivery or implementation. Summative evaluations typically occur at the end of a program (ex-post evaluation) with a retrospective and holistic scope that assesses all program aspects including delivery, activities, outcomes and impacts. This form of evaluations helps make judgements of the program's overall success in terms of its effectiveness (e.g. were the intended outcomes achieved?), efficiency (e.g. were resources used cost-effectively?) and appropriateness (e.g. was the program a suitable way to address the needs of the group targeted?).

In this report, we focus on methods of ex-post evaluation, which examine the actual outturn of a policy, programme or project against its projected outturn. Ideally, these approaches also assess the outturn against a "no policy" (no project) scenario, by establishing a counterfactual against which to measure impact. Evaluation findings provide valuable feedback for future policy design, and are a distinctive learning contribution to inform the policy cycle and, in the case of the instruments promoted by CRP Livestock, the wider implications on the design of investment strategies for sustainable development. To distinguish the effects of an intervention from the effects of external factors and to understand how impact is achieved (or not) is a matter for ex-post impact evaluation. In this regard, two approaches are interlinked in establishing impact of an intervention: counterfactual approach and theory-based approach. The two approaches are interrelated in any assessment of impact and will typically feature as ex-post evaluation questions.

The counterfactual approach is founded on the question "*Did it work*?" That is, the main question driving counterfactual assessments in evaluation is if an intervention has had an effect and if so, the extent of this. This question cannot be separated from why an intervention works as a theory of change approach is necessary to help decide which changes should be attributed to a cause. The main methods recommended draw on econometric approaches and involve the use of techniques such as difference-in-difference, discontinuity design, propensity score matching, instrumental variables and randomized controlled trials. To allow application of counterfactual approaches, accurate baseline and follow up data is required both for the beneficiaries (supported group) and non-supported group.

The second approach is the theory-based approach, which is founded on the question "*Why did it work?*" This approach allows decision-makers decide on what course of action to implement through identifying why a set of effects has been produced. It is presented as a narrative and is largely based on qualitative methods and helps strengthen or reconstruct an intervention logic or logical framework. Aspects beyond measurable effects to identify why an intervention produces effects, how, for whom and under what conditions and intended/unintended effects. Although, theory–based evaluations are not logical frameworks, without a clear logical framework in place from the outset, a theory-based evaluation faces multiple risks as it tries to reconstruct a logical framework and draw lessons for the future. How to deliver a theory-based evaluation is not explicitly dealt with in this report.

In the next sub-sections, we summarize the main findings of our scoping review of *ex-post* tools, methods, and approaches that are commonly used in the literature to assess the impact of research and development investments in general, particularly those that may be relevant and suitable for CRP Livestock. We first describe each of the identified methods and approaches including their purpose and application, and their advantages and disadvantages. Next, we discuss the constraints under which each method can be used and their applicability to CRP livestock interventions.

1.1. Methods of ex-post impact evaluation

A range of IE methods exists in literature and have been applied in practice for ex-post assessment of interventions and programs and could be deployed to assess CRP Livestock interventions. The following discussion introduces six ex-post IE methods and approaches in brief and considers their strengths and weaknesses in relation to their application to the CRP Livestock context. These methods are broadly categorized into experimental (i.e. randomized evaluation) and non-experimental methods including matching methods (propensity score matching), double-difference methods, instrumental variable methods, regression discontinuity methods and endogenous treatment and control function methods. It is important to mention that there are other structural and empirical approaches not discussed in this report.

1.1.1. Randomized evaluation method

Randomized evaluations, commonly referred as the randomized control trials (RCTs), encompass the random allocation of eligible units (e.g. individuals, communities, business, household, etc.) to a "treatment group" that benefited from an intervention and to a "control group" that did not benefit from the same intervention (Khandker et al., 2010; Rogers et al., 2015). The beneficiaries and non-beneficiaries of the intervention have the same pre-intervention characteristics. This method gives program implementers and managers a fair and clear rule for assigning limited resources equally amongst eligible populations. The use of RCT method to evaluate impacts of social programs has increased substantially in recent years, which comes with several advantages. First, since the objects are randomly selected into the treatment and control group, RCTs avoid selection bias. If, for example, people who share a common characteristic had been more prone to self-select themselves into the treatment group, selection bias would be a problem. RCTs however, give all eligible prospective beneficiaries the same chance of being selected into the intervention. Also, RCTs provide clear impact pathways, as a major strength of this methodology is the straightforward investigation of cause–effect relationships that it enables. This is due to the fact that confounding factors and bias are practically absent. Conclusively, RCTs are commonly referred to as the gold-standard when studying causal relationships. Nonetheless, there are also certain drawbacks to take into consideration when using randomized evaluations methods. An obvious one is budget limitations, due to logistics, that sometimes prevent programmes and interventions from reaching the entire population of interest. Power calculation and validity might require large samples size and that the experiment is conducted at multiple sites, which might be conditions that are cumbersome for the investigators to bring about. The timeframe of the experiment can also be tricky to manage, as capacity restrictions sometimes prevent the roll-out of programmes and interventions to eligible participants at the same time, not to mention the fact that it might take a long time for the desired effects from the experiment to occur.

1.1.2. Matching methods

The matching method, precisely propensity score matching (PSM), generates a comparison group from control units by matching treatment units to one or more observations from the controlled sample, based on observed features. This method is applied when involvement in the intervention is voluntary. Treated units are matched to control units that have similar propensity score. The PSM is based on two key conditions: (i) conditional independence, *i.e.*, unobservable factors doesn't influence programmed participation, and (ii) substantial common support in propensity scores across programme participants and non-participants. PSM estimation demands data from both treatment and control groups. The major advantages of matching methods are that they can be applied ex-post, even when there is no baseline data, and to a dichotomous treatment if there is adequate data. Potential drawbacks however, stem from the fact that PSM solely depends on the matching on observed factors. Hence, if the selection of units is influenced by unobserved factors, PSM will produce biased results in an ex-post single assessment (King and Nielsen, 2016). Another troublesome factor to consider is the difficulty of actually identifying credible groups that look alike, which is essentially the cornerstone of the PSM.

1.1.3. Double-difference methods

Unlike the PSM discussed above, double-difference (DD) methods presume that there exists unobserved heterogeneity in selection into the programme or intervention. However, DD assumes that the unobserved factors are time invariant. Hence, the DD estimator depends on a comparison of treated and controlled units prior to and after the intervention (Khandker, Bakht, and Koolwal, 2009). This method thereby addresses impending causes of selection bias by basically comparing the treatment and the control group in relation to changes in outcomes over time compared with the perceived outcomes projected during the pre-intervention stage (Khandker et al., 2010). Therefore, an advantage of using DD methods is that they ease the assumption of conditional exogeneity or selection only on observed characteristics. At the same time, DD methods offer a simple and intuitive technique for addressing the problem of unobserved factors. The main weakness of DD in this specific case however, *i.e.*, when assessing the impact of CRP Livestock interventions, is that the assumption of time-invariant selection bias does not hold for several of the targeted interventions when it comes to these emerging and rapidly changing economies.

1.1.4. Instrumental variable methods

Instrumental variable (IV) methods permit the unit (e.g. people, communities, business, household, etc.) selection into interventions to be endogenous. Hence, the IV methods deal with the identification of one (or several) relevant instrument(s) that is highly correlated with selection and placement into interventions but uncorrelated with unobserved factors that influence the intervention outcomes. Specifically, the IV solution is to isolate the variation in the explanatory variable that is unrelated to the confounding variable(s) in question. The variable which does this “isolating” is referred to as the instrumental variable (IV). The main advantage of this method is that when valid instruments are identified, the experimenter is able to control for observed and unobserved biases. Hence, in theory, the IV approach is practically bulletproof. Another advantage is that the experimenter only needs some of the variation in the explanatory variable, so the requirements on the instrument are weaker than those on the confounding variables that otherwise would be included in the regression as controls. However, there are some main drawbacks of using IV methods as well. An obvious one is the difficulty in finding valid instruments, considering that several factors which influence selection into the intervention usually influence outcomes in a certain. The IV method also produces latent average treatment effects, which may sometimes be hard for policy makers and other stakeholders to understand.

1.1.5. Regression discontinuity method

The regression discontinuity (RD) method is applied when there exists a benchmark for a given unit to be eligible for selection into the intervention. For example income status; if a certain amount is set as a threshold for an individual to partake in an intervention (Thistlewaite and Campbell, 1960). This method assumes that units that are closer to either side of the benchmark are very much alike for untreated units from the intervention to be an effective control unit. The difference in outcomes between the units closer to either side of the benchmark is attributable to the intervention. The benchmark should be unique to the intervention. An advantage of using the RD method is that it better addresses unobservable factors compared to other non-experimental matching approaches, since there is no selection bias as long as there exists a solid and fixed benchmark that forces units into selection regardless of any other characteristics.

RD also uses administrative data to a great extent and thereby minimizes the need for additional data collection. Furthermore, thanks to the nature of the method, the results can easily be illustrated graphically in a scatterplot – as one expects to see the trend line "jump" around the benchmark. These scatterplot graphs generally make the RD-results more compelling to look at. In fact, it is even consider a bad practice to report RD regression results without the corresponding graphs. This is due to the fact that if the results are clear in the regressions they should be clear in the graph too. Hence, failure to show the graph can actually raise suspicion about the reliability of the findings. A potential drawback of the RD method is that there must be a clear benchmark for assigning units and that the impact assessment is only possible for the population close to the benchmark. This is due to the fact that the farther away from the benchmark, the more will other factors that differs between the observations inhibit the possibility to draw any conclusions regarding the cause and effect relationship. This stems from the fact that the RD-method relies on the assumption that objects close to the benchmark, while on both side of it, share the same characteristics aside from being on different sides of the cut-off.

1.1.6. Endogenous treatment and control function methods

Endogenous treatment and control function methods follow two stages. In the first one, participation in an intervention is captured as a dichotomous variable; for instance, participation in microfinance or a food security programme. A binary probit regression is then estimated and from this regression, predicted values are created. The predicted values are used in the second stage, where the outcome variables of interest are regressed on the predicted participation in the intervention as well as other variables that can influence the outcome variables (Heckman, 1976). An advantage of using this method is that it can often be used as long as there is appropriate data. Also, the method is able to detect the presence of selection bias. Another advantage of endogenous treatment and control function methods is that they generate average treatment effects of interventions that are of great interest to policy audience. In order to apply these kind methods, at least one instrument is required. However, as for the IV, it is many times difficult to find valid instruments, which most definitely has to be considered a drawback of the endogenous treatment and control function methods.

1.2. Methods for synthesizing evidence from impact evaluations

A range of methods in the literature have also been applied in practice to synthesize evidence from developmental interventions with particular focus on resources. These could be deployed to assess CRP livestock interventions. The following sub-sections briefly present ten IE methods considering their advantages and drawbacks in relation to their application to the CRP Livestock context. It is important to mention that some of these methods are applied for monitoring and evaluations i

1.2.1. Cost-benefit analysis/ cost-effectiveness analysis

Both Cost-benefit analysis (CBA) and cost-effectiveness analysis (CEA) methods are employed to ascertain whether or not the intervention or programme costs can be justified by its outcomes and impacts. These methods have been applied by development organizations for assessing whether the costs of development programmes are greater than or less than the benefits (Belli et al., 2000). Although, both methods are used for drawing similar conclusions, CBA quantifies both inputs and outputs in monetary values whereas CEA quantify inputs in monetary values and outcomes in non-monetary quantitative values with the use of a common outcome metric (World Bank, 2004; Rogers et al., 2015). The CBA and CEA methods are applied as ways of helping project funders, implementers and managers to allocate resources efficiently– by systematically estimating the strengths and weaknesses of alternatives that determine which options provide the best approach in order to achieve benefits while preserving resources. Essentially, they provide a basis for comparing decisions or investments by taking into account the total expected cost of each option and weighing it against its total expected benefits. In that sense, CBA/CEA methods can be applied when evaluating either completed or potential courses of actions, i.e. both ex-post and ex-ante. The methods thereby evaluate the value against the cost in order to identify the interventions or programmes with the highest rates of returns. There are several advantages of using CBA and CEA methods. One is that they provide better opportunities to draw well founded conclusions when assessing the efficiency of interventions, as both methods make clear economic assumptions about development interventions and impacts. This also make the CBA and CEA methods suitable for convincing programme funders and stakeholders about the viability of their funded projects. Potential drawbacks concerning the CBA and CEA methods are that they can become rather technical and therefore require sufficient pecuniary and human resources. It can also be difficult to acquire the needed cost and benefit data for analysis and as such results from these methods may be much reliant on assumptions. In addition, CBAs in particular may under- or overestimate the impact of an intervention, especially when benefit streams interact.

1.2.2. The logical framework method

The logical framework is a useful tool for engaging programme administrators, funders, managers and other partners in clearly defining project, policy or intervention goals and planning of project activities. This framework is also used at the implementation stage of an intervention to assess growth and to ascertain if there are corrective measures that needs to be taken (GTZ, 1997; World Bank, 2000). In addition, this framework helps identify the impact pathways or links "programme logic". The programme logic follows a chain: inputs, processes, outputs, outcomes, and impact. At each stage of the chain, the framework identifies performance indicators and factors that might hinder the achievement of the project, policy or intervention goals (Romano and Wolf, 2005; World Bank, 2000). The Logical Framework is used for enhancing the quality of projects, programmes, and intervention or policy designs by enabling conceptualization and planning of multifaceted programme and intervention activities. Since it provides a solid basis for reviewing, monitoring and appraisal of programmes, interventions and policies, it thereby supports the preparation of comprehensive working plans. An advantageous feature of the Logical Framework methods is that key investors and partners become involved already in the preparation and monitoring of programmes and/or interventions. The method thereby allows the programme partners, including funders, decision-makers, managers and other stakeholders, to raise important questions and scrutinise assumptions and threats to programmes and interventions. These generous participation barriers, even though beneficial in several aspects, come with some natural drawbacks as well. For example, they require routine updates along the implementation stages of the programme and might require training and follow-up of participants and implementations. Also, if handled strictly, it might actually suppress originality and novelty.

1.2.3. Public expenditure tracking surveys

Public expenditure tracking surveys (PETS) track the movement of public resources (e.g. research funds) with the aim of finding out the degree to which public resources truly reached the targeted beneficiaries. PETS take into account the means, amount and times of delivering resources to the designated units, and are usually applied in surveying big programmes or interventions in terms of their delivery features. PETS are thereby used for detecting anomalies in project delivery quantitatively and for signalling project interruptions. They also function as an important tool to identify corrupt practices in the vast nexus of funders and beneficiaries. Using PETS therefore comes with the advantage of ensuring accountability in a programme implementation. It also helps in identifying managerial and administrative barriers in the flow of project funds. Potential drawbacks are that access to accounting data from state agencies and departments can be difficult to obtain and that the implementation. Additionally, the management of the tracking system generally is an expensive process.

1.2.4. Participatory methods

Participatory methods allow project partners and investors to actively participate in decision-making regarding a programme or intervention as well as the implementation strategies. This allows the stakeholders to have a sense of ownership in the monitoring and evaluation of programmes and interventions (Guijt, 1998). The widely used participatory methods include stakeholder analysis, beneficiary analysis, participatory rural appraisal and participatory monitoring and evaluation (see more details at from Guijt, 1998). The purpose of these methods is to improve and sustain programmes, interventions or policies by appraising and detecting eventual implementation difficulties among them. Another essential function is to empower deprived people through knowledge and skill sharing during their participation. There are several advantages of using participatory methods. First, they enable the studying of important issues by including key stakeholders in the programmes, interventions or policies process. They thereby enhance collaborative ownership of these programmes, interventions or policies as well. Second, participatory methods improve indigenous knowledge, administrative capacity, and skills by including the domestic population in the interventions. Third, these methods offer timely and credible evidence for policy decisions. The natural drawbacks of using participatory methods however, is that it is laborious and time consuming to constantly involve all relevant stakeholders. In addition, some of these stakeholders might even seize the opportunity to try to pursue their own agenda.

1.2.5. Rapid appraisal methods

Rapid appraisal methods, as the name implies, are fast and cheaper means of gathering opinions, ideas and feedback of participants, key informants and other stakeholders about the impacts of particular programme and intervention, for the purpose of answering to project funders' or donors' demand for evidence of impact (Kumar, 1993; USAID, 2010). Widely used rapid appraisal methods include key informant interview, focus group discussion, community group interview, direct observation and mini survey (Kumar, 1993; USAID, 2010). At the programme or intervention level, these methods can offer quick information for programme implementers and funders and also provide detailed and contextual understanding of quantitative findings. Rapid appraisal methods are advantageous since they are quick to both implement and conduct, which implies that they are also cheap. Consequently, the methods are flexible as well and can therefore easily be used to search for new ideas. A drawback of the rapid appraisal methods is that findings from them are location specific and can therefore not be generalized or implemented anywhere. This goes along with the fact that findings from these methods have weaker validity, reliability and more credibility issues relative to formal surveys.

1.2.6. Formal surveys

Formal surveys employ standard sampling approaches to collect data from selected units (e.g. people, households, communities, business, etc.). Formal surveys collect and compare data from large samples in a particular target population (Sapsford, 1999). Some of the common formal surveys include Living Standards Measurement Survey—LSMS, Core Welfare Indicators Questionnaire (CWIQ), Client Satisfaction Survey and Citizen Report Cards. These surveys are used for baseline or pilot studies to scope future impact comparison, for matching different clusters at a particular period of time and/or the same clusters over time, and for describing circumstances in a given region, city, community or group. The purpose of formal surveys is thereby to provide relevant inputs for proper impact assessments of programmes and interventions, and to measure welfare levels and strategies.

Hence, an advantage of using these surveys is that their findings usually involve large samples that can be generalized or applied to a larger share of the population. Also, quantitative findings can be obtained for the sampled respondents and distribution of impacts. Some potential drawbacks of using formal surveys are that the results are usually not accessible for a long period of time and that larger survey data generally requires advanced technical skills and expert knowledge to be properly analysed. This goes along with the fact that formal surveys like LSMS are costly and time demanding to conduct, which is partially because some types of information are very hard to get from formal interviews.

1.2.7. Value for money assessment

Value for money (VfM) assessment is an evaluative approach that answers the question about how well resources are being used and whether this usage is justified (King, 2017). Impact evaluation according to VfM intends to clarify whether an investment is well managed and worthwhile on the basis of observable features of programme delivery, and agreed definitions of what 'good' performance would look like (King & OPM, 2018). Moreover, VfM seeks to find the right balance between economy, efficiency and effectiveness, and cannot be assessed through only one of these dimensions in isolation. Reducing the costs of inputs and making efficiency savings can either support or undermine value for money (King & OPM, 2018). Traditionally, results from VfM assessments focus on three Es; Economy, Efficiency, and Effectiveness. However, a fourth E (i.e. Equity) has gained more attention in recent years (King & OPM, 2018).

Advantages of applying the VfM assessment framework is that it stretches beyond merely cost-cutting or efficiency. Essentially, it provides insights on the four Es; Economy, Efficiency, Effectiveness and Equity. For instance, in terms of effectiveness, an intervention may can be judged as less effective because of a small cost saving, value for money is reduced. Similarly, while an intervention may be very cheap and run efficiently, if it does not achieve results, it is definitely not value for the money. Hence, when evaluating the impact from the VfM standpoint, the quality of the outcomes is fundamental to understanding whether something is actually providing value (Jackson, 2012). There are, however, several drawbacks of the VfM framework. The OECD Development Co-operation Directorate (2012), claims that one complicating factor is the fact that the actual VfM assessment is harder in the development context than elsewhere. This is because, in some developing countries, the availability of reliable information, notably statistics, is often of too poor in terms of quality to make any reliable assessment. Another drawback is that there is a lack of agreement on value for money for whom, of what and by when (Jackson, 2012). In particular, in international development, the question of value for money from whose perspective is important since the immediate beneficiaries and funders are not the same people. For example, there is a valid concern that what value for money means for a donor is not the same as what it means for a partner country or for an individual beneficiary.

1.2.8. Multi-criteria analysis

Multi-criteria analysis (MCA) is used for comparative assessments of alternative projects, programmes and interventions. The method aims to assist decision-makers in integrating different options into a prospective or retrospective framework. Hence, MCA enables the possibility to have several criteria being taken into account simultaneously in a complex situation, and the results are usually directed at providing operational advice or recommendations for future activities. Due to the nature of MCA, participation of the decision-makers in the process is a cornerstone of the method, and the decision-makers come together with the objective to produce a single synthetic conclusion or, on the contrary, to produce conclusions adapted to the preferences and priorities of several different partners. The MCA resembles cost-benefit analysis. The purpose of the MCA method is to structure and combine the different assessments required for decision-making, which is made up of multiple choices. In short, MCA is thereby applied in order to provide a single framework that, at the ex-ante stage, makes it possible for the decision-makers or evaluators to get a range of performance-related information. At the strategic level though, in which it is less likely that targets will be fully quantified, then a logical framework is necessary and MCA might have a key role to play in providing a clear organising framework to evaluate performance across multiple objectives or types of potential intervention.

MCA can generally be concluded in 5 steps: *1.* Definition of the projects or actions to be judged. *2.* Definition of judgment criteria. *3.* Analysis of the impacts of the actions. *4.* Judgment of the effects of the actions in terms of each of the selected criteria. *5.* Aggregation of judgments. These steps will naturally involve an inventory of the planned or implemented actions, or the elements on which the comparative judgment will be made. A clear advantage of the MCA technique is that it brings forth the individual opinions and values of several agents in a quantitative way. MCA is therefore well aligned with the way in which partnerships work in general, as it outlines areas of consensus in which the partners agree, and areas of dissension where the interventions in question might be considered successful for some and unsuccessful for others. Hence, MCA might help partners to compromise on issues or establish a coalition of views. The method is also advantageous since decision makers most often prefer to be involved in the process through a relatively simple technical framework, such as the MCA. A potential drawback could be that, in the strict sense of the term evaluation, MCA is seldom used for purposes outside of decision-making. This is because specific problems of implementation may limit the use of this method, or require the presence of experts. In addition, this method is not always used in an interactive way as possible, and therefore tends to stiffen criteria that in reality are fluid.

1.2.9. Spatial impact analysis

Spatial (territorial) impact analysis is applied when assessing impact at the area-based dimension – with area-based regeneration within cities usually being the main objective. In short, this type of analysis is thereby necessary when the geographic area at which impacts are supposed to take place has to be considered, with the purpose of judging the performance of the operation in question. Whichever the evaluation paradigm being, a key performance measurement is the choice of spatial area within which the impacts are being assessed. This choice is highly important for all kinds of interventions, since territorial focuses and spatial dimension are of pivotal relevance when judging performances. Hence, the choice of area will usually depend on the objectives of the initiative in question. For example, some initiatives may intend to solve certain issues at the city-wide level, while others may instead operate across a larger metropolitan region. Therefore, an advantage of this method is that the selection of the most suitable spatial area for impact evaluation has implications for every key component that bears relevance in the impact measuring process. In Methodologies for Assessing Social and Economic Performance in JESSICA (2013), some potential drawbacks of the spatial impact analysis are also presented. These drawbacks generally lie within challenges associated with assessing inclusion at the city level. Such challenges might be issues concerning data availability **–** both financial and non-financial, the quality of the data and its overall reliability and comparability. Hence, when assessing non-financial impacts it might be necessary to design and deliver sample surveys to overcome these issues concerning data availability. Also, displacement tends to be high in relation to retail activity for example. Here, the vaster the spatial area, the larger the level of expected displacement effects, *ceteris paribus*. However, the opposite relation applies to leakage (the proportion of outputs or outcomes that benefit those outside the targeted spatial area). Thereby, leakage tends to decrease with the width of the spatial area.

1.2.10. Outcome mapping

Outcome mapping (OM) unpacks a programme or an intervention’s theory of change and gives a framework for immediate data collection, rudimentary changes that resulted to long-term and more developmental change. It also allows for the credible evaluation of the program’s and intervention’s impact through ‘frontier partners’. OM tools can be applied as stand-alone or together with other monitoring and impact evaluation methods. This method can be used for identifying agents (e.g. institutions, society, groups or individuals) needed to effect developmental change. It is used in planning and monitoring welfare and developmental change and the policies to support those changes. It is also a useful tool for observing internal activities of programmes or interventions to keep them effective as well as for creating an assessment framework to inspect specific issues relating to the programme or intervention. Some advantages of using OM are that: it is a rigorous method that is applicable to different contexts. It provides detailed insights and understanding of impact path ways. It also enhances efficiency and offers realistic and accountable reporting. Some drawbacks of using OM are that: this method needs experts to facilitate the evaluation process. OM requires huge budget and much time. It also needs a “mind change” of individual and institutional paradigms.

1.3. Framework for selecting appropriate designs and methods for impact assessment

In this sub-section, we discuss a framework for appropriate designs and methods for impact assessment of development interventions, programs and policies. In addition, we discuss how the available methods can best be assigned to specific types of development interventions, programs and policies. Figure 1 presents a framework proposed by Stern et al. (2012) for designing impact assessment. As shown in Figure 1, there are three key elements in the framework:

1.3.1. Resources and constraints

Choosing an appropriate impact assessment method depends on the existing resources and other constraints. These comprise of availability of existing and reliable evidence from earlier research, evaluation and monitoring and ability to carry out further evaluation in terms of funds for external evaluators or evaluation team, including contractual and procurement expenditure; funds to cover internal cost including cost of internal staff, project managers and administrators involved in the evaluation work; available experts with the required knowledge and skills needed for evaluation assignment; policy and implementation gap; time availability; political influence on resource availability and termination of project; and, organizational constraints including ethical issues, standards, norms, policies and processes leading to reliable evidence.

1.3.2. Nature of what is being evaluated

The nature of the program being evaluated plays an important role in selecting an appropriate method. The features of the development interventions, programs and policies are usually described in a theory of change. The theory of change describes how the interventions and programs inputs (e.g. funds, people, products, and changes in regulatory or policy environment) contribute to short and intermediate outcomes that yielded the anticipated and unanticipated potential impacts (Carvalho and White, 2004). The theory of change helps to identify important assumptions and external factors that can potentially affect the degree to which outputs will lead to short and intermediate outcomes and anticipated and unanticipated impacts. It also gives a structure for building an evidence-based description of the value and impact of interventions, programs and policies.

The inclusion of theory of change in the design of interventions, programs and policies improves the quality of the design and implementation of the impact assessment. In addition, the theory of change identifies coherent gaps, significant risks and provides a common understanding for stakeholders who may be unfamiliar with the technical ways of presenting impact findings. The stage of the interventions, programs and policies in the project lifecycle plays critical role in choosing appropriate design and method for impact assessment. It is very important to determine whether the interventions, programs and policies to be evaluated is simple, complicated or complex.

- **Stage of the project**- The stage of the intervention or program has significant implication on the type of design and method to be used. It is important to ascertain whether the intervention program is at roll-out or pilot, middle or final stage. For new interventions, programs and policies program, the researchers can generate a control group. Creating a control group is very complex and often not possible for interventions and programs that have already been widely executed or at its final stages.
- **Visibility of outcomes and impacts-** In selecting appropriate designs and methods, it is relevant to gather evidence on the nature of outcomes and impacts. Information on activities and impacts are needed to differentiate between execution and theory failures. For some interventions and programs, both outcomes and impacts are easily observable and in such instances it is easy to create reliable measures for intended outcomes and impacts. For some interventions and programs where outcomes and/or impacts are not visible, the researcher must combine different indicators. For some interventions and programs with standardised procedures (e.g. like evaluating applications), the evaluator can create a standard for measuring the quality of the program execution in a consistent manner. For some interventions and programs that involve custom-made implementation (e.g. particular advice to smallholder farmers or agri-firms), the evaluators will need to use expert judgements for assessment. The impact evaluation of interventions and programs that have transformational impacts that are not readily reversed (e.g. transfer of new knowledge) can be performed at any period, given that impacts can be measured at any time (Rogers et al., 2015). The impact evaluation of interventions and programs with crumbly impacts that can be simply reversed is very complex in terms of finding the appropriate time for evaluations.

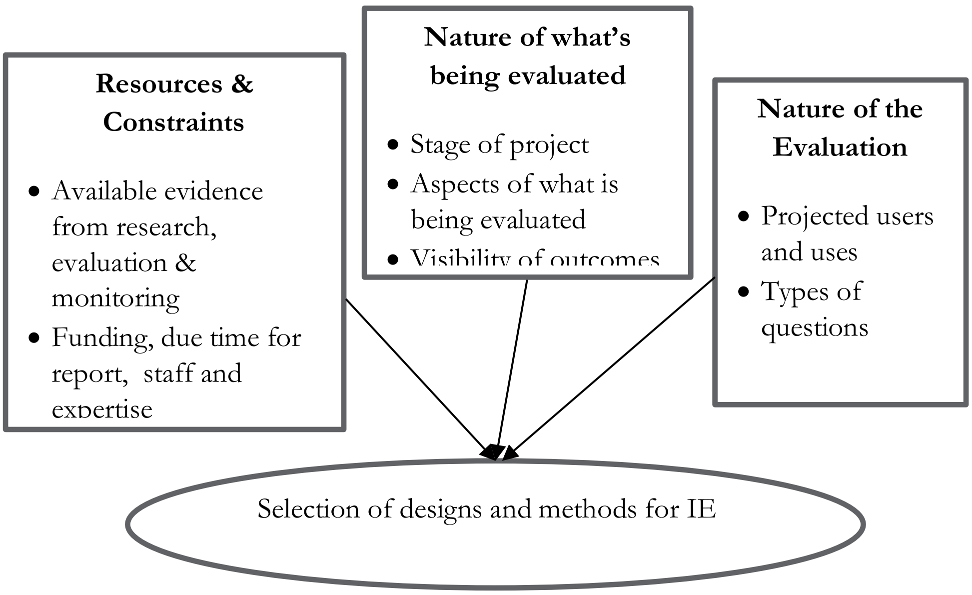


Figure 2. Framework for selecting appropriate methods and designs.
Adapted from Stern et al. (2012) and Rogers et al. (2015)

- **Aspects of interventions, programmes and policies being evaluated-** In choosing appropriate designs and methods, it is important to categorise the interventions and programs into three typologies consisting of simple, complicated or complex (Stern et al., 2012; Rogers et al., 2015; Stacey, 1992; Kurtz and Snowden 2003). These typologies are very relevant for planning and analysis of impact assessment (Guijt, 2008; Rogers, 2011). These typologies does not necessarily apply to the entire intervention but can be used to categorize aspects of the intervention or program. 'Simple' typology of interventions and programs are usually standardized (e.g. particular products, technique or process). 'Complicated' typology consist of multiple dimensions, usually a portion a larger multi-dimensional program, or work separately as part of a bigger causal package (e.g. participants in a given implementation setting,). 'Complex' typology is dynamic and developing aspects of programs implemented as a response to emerging needs and opportunities (Rogers et al., 2015).

1.3.3. The nature of the evaluation

The nature of the evaluation depends on projected uses and users and the types of evaluation questions. The planned uses of the impact assessment raises different evaluation questions. Various planned users have different projected uses and what they consider as reliable evidence.

1.4. Applicable methods & designs for evaluating impact of CRP livestock intervention's

So far it is clear from this scoping review that there are a variety of approaches, designs and methods for analysing and synthesizing impacts of programmes and interventions. Since evaluators cannot adopt all the existing methods, a criteria is adopted for selecting an appropriate method (s) for the CRP livestock intervention. The criteria employed is based on the aim of the CRP assignment, the stage of the intervention, available data, budget and time allocated for the assignment. To begin with, the aim of the study is to undertake an IE study to assess the return on investment of the activities and interventions implemented by CRP Livestock over the last five years in terms of their efficiency, effectiveness and the extent to which they have achieved programs outcomes. In line with the objective of the study, an appropriate method should capture the efficiency and effectiveness aspects of the intervention.

In terms of stage of the CRP intervention, IE evaluators were invited at the end of the intervention. Thus, they were not involved in the intervention from the onset and as such does not have a baseline data required for rigorous impact assessment using counterfactuals. In addition, the time and budget allocated for the assignment does not permit evaluators to conduct a formal survey to collect data from treated and control groups and further analyse the impact. Hence, we placed the various reviewed methods in the context of their common applicable stages in the project cycle, constraints under which they can be applied for synthesizing evidence and their applicability for CRP livestock interventions assignment.

Table 1: Applicable methods and designs to CRP livestock intervention's assignment

Methods for synthesizing evidence from monitoring and impact evaluation	Most applicable stages & scenarios			Constraints under which IE methods & designs are used			Applicability to CRP livestock interventions
	Start	Monitoring & control	End	Time	Data	Budget	
Cost benefit analysis/ cost-effectiveness analysis							√
The logical framework							×
Public expenditure tracking surveys							×
Participatory methods							×
Rapid appraisal methods							√
Formal surveys							×

Outcome mapping							×
Value for money							√
Multi-criteria analysis							×
Spatial impact analysis							×

Table 1 presents the different reviewed methods and their applicability to the CRP livestock intervention's assignment. As indicated in Table 1, methods and designs such as the logical framework, public expenditure tracking surveys, participatory methods, formal surveys, spatial impact analysis and multi-criteria analysis cannot be applied in this CRP livestock assignment due to time, data and budget constraints under which this study is to be conducted. Thus, IE evaluators were invited at the end of the intervention and are tasked to complete the evaluation within a tight time frame coupled with no baseline or panel data on treated and controlled groups as well as limited budget. Specifically, the logical framework is not applicable to this CRP assignment because the intervention is ending. This tool is useful tool for engaging programme administrators, funders, managers and other partners in clearly defining project, policy or intervention goals and planning of project activities. Public expenditure tracking surveys is not relevant or applicable in this CRP assignment as it's mostly used in the public sector to track movement of public resources and as such does not clearly fit in the aim of this study. Since the CRP intervention is ending, the participatory methods will not be relevant or applicable for this assignment because it's too late to reconsider project partners and investors to actively participate in decision-making regarding a programme or intervention. Formal surveys will now be applicable in this assignment since no surveys were conducted to collect data from selected units (e.g. people, households, communities, business, etc.) during the intervention years. Multi-criteria analysis is also not applicable or relevant to this CRP assignment because it is mostly used at the ex-ante stage of projects, programmes and interventions. In addition, since this assignment does not focus on specific geographical location, spatial impact analysis is not relevant to the CRP assignment.

In addition, given that the time-frame doesn't allow for formal surveys for collecting data on treated and control groups, rigorous impact evaluation methods such as the randomized evaluation, matching (propensity score matching, double-difference, instrumental variable, regression discontinuity, endogenous treatment and control function methods cannot be conducted for this assignment. Given the study's aim, stage of the project and the constraints under which the impact evaluation is to be conducted, designs and methods such as cost benefit analysis/cost-effectiveness analysis, rapid appraisal methods and VfM can be applied in this assignment. As discussed above, and in line with the objective of the CRP study, cost benefit (CBA) and cost-effectiveness analysis (CEA) is relevant and applicable to this study these take into account efficiency and effectiveness of an intervention and can be applied both ex-post and ex-ante assessment. These methods are known for convincing programme funders and stakeholders about the viability of their funded projects. However, it is worth mentioning that it will require administrative data on cost and benefit, some of which are difficult to obtain. Secondly, considering the time, budget and data constraints, rapid appraisal method is also relevant and can be applied in the CRP assignment because as the name implies, it is a fast and cheaper method that can be used to gather programme and intervention impacts from the viewpoint of participants, key informants and other stakeholders. It is also useful for answering to project funders' or donors' demand for evidence of impact. It is important to mention that outcome mapping (OM) will be relevant for unpacking the CRP intervention's theory of change and impact. Finally, VfM is quite relevant and applicable for this study as its assessment takes into account economy, efficiency and effectiveness, all of which are key aspects of the aims of the CRP assignment. Although, these three methods are relevant and applicable to the CRP assignment, we intend to dwell much on value for money in combination with aspects of rapid appraisal approaches such as expert or key informant judgments and weighting for key outcomes and impact areas.

2. Proposed VfM assessment framework for CRP Livestock

In light of the findings of Section 1 of this study, and based on discussions and consultations with the CRP Livestock team regarding the nature of the program interventions and research activities, the study team concluded that a VfM assessment framework is the most relevant and applicable analytical framework for the CRP program. The most prominent reason for our selection of VfM assessment is that the CRP Livestock community generally aims for performance criteria that differ a lot from those in other areas of public spending: namely in ways where the mere amount spent tends to overshadow the question of what the funds actually achieve. Another reason is that CRP Livestock is pressed to demonstrate the value for money of their work in front of their funders, and VfM assessment can help achieve this objective because it takes into account economy, efficiency and effectiveness, all of which are key aspects of CRP Livestock research investments. In this section of the report, we first provide a more detailed description of the VfM approach for impact evaluation, and we then propose a VfM-based framework for rapid ex-post impact assessment of CRP Livestock.

2.1. Brief description of the application of the VfM assessment approach

To apply a VfM framework a development context in a useful way, it is important to highlight that VfM is not the same as cost-cutting or efficiency (Jackson, 2012). That is, VfM is not synonymous with either only economy (i.e. reducing the cost of inputs) or efficiency, but is blend of both (and more). In essence, the VfM framework seeks to capture the relationship between cost and value. The cost element usually represents the cost over the life time of the project to deliver the associated value, including the costs of managing the associated risks in doing so. Value comprises the quality and quantity of service or performance level over the same period. This relationship between value and cost can be presented conceptually in the form of Figure 3 below, in which different options represent different combinations of performance and cost. The general stages involved in VfM framework design include specifying a theory of change, identifying VfM criteria, developing standards for each criterion, and determining the evidence necessary to support well-informed evaluative judgements against the criteria and standards.

- **Step 1: Theory of change (ToC):** A ToC explains how activities are understood to produce a series of results that contribute to achieving the final intended impacts (Rogers, 2014). One of the functions of a ToC is to assist in the identification of criteria, standards, and indicators that are relevant to the programme's results chain. By following the sequence shown in Steps 1–4, starting with a ToC before moving on to the development of criteria and standards and then the selection of metrics and methods, it is ensured that the values embedded in the criteria and standards reflect the programme theory, and that the metrics in turn reflect the content of the criteria and standards. This helps ensure that the indicators we use are valid, cohere with the programme theory, and provide relevant evidence to support contribution analysis and evaluative judgements. Typically, a theory of change will be developed during the inception phase of a project – so it makes sense to integrate the development of the VfM framework with this activity. A good ToC is explicit about resources and inputs and how they are supposed to be used to produce outputs and outcomes – so it should already contain all of the information needed to support the identification of VfM criteria and standards.

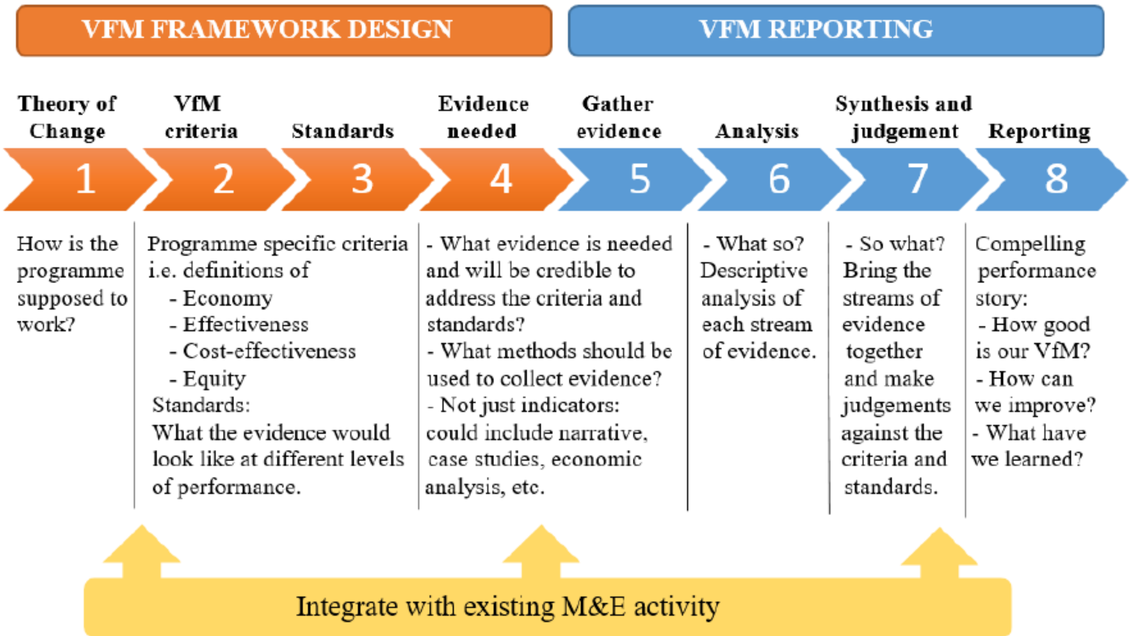


Figure 3. Overview of evaluation-specific approach to VfM
Source: King & OPM (2018)

- **Steps 2 and 3: VfM criteria and standards:** The performance of complex programmes should not be judged solely on the basis of indicators, devoid of any evaluative judgement. Well-reasoned judgements of VfM are required. Criteria and standards provide an agreed, transparent basis for interpreting the evidence and making these judgements. In this context, criteria of merit or worth are selected dimensions of performance that are relevant to a particular programme and context. They describe at a broad level the aspects of performance that need to be evidenced to support an evaluative judgement about VfM. When using the Four E's (economy, effectiveness, equity, and cost-effectiveness) and efficiency as the basis for VfM assessment, criteria need to be defined for economy, efficiency, effectiveness, equity, and cost-effectiveness. Performance standards also need to provide defined levels of VfM (i.e. excellent, good, adequate, and poor) for each of the criteria. Criteria and standards should reflect the key inputs, activities, deliverables, outputs, intermediary and long-term outcomes, and impact articulated in the theory of change. In particular. The VfM is about finding the right balance between the these dimensions (the four E's) as follows:

    - **Economy**: A measure of what goes into providing a service: "Are we or our agents buying inputs of the appropriate quality at the right price?" (Inputs might be things such as staff, consultants, raw materials and capital that are used to produce outputs). Unit costs are typically used as an economy measure. When identifying unit costs, the whole life costs of inputs, such as the direct and indirect costs of acquiring, running and disposing of assets or resources, should be considered. Hence, measuring the economy of a project/partner is all about input costs and what drives them. However, a common misunderstanding about VfM is that it is all about economy – which is definitely not the case, but accessing information on economy is relatively easy through the budget and hence often becomes the de facto focus of VfM assessment.

    - **Efficiency**: A measure of productivity, in other words: "How much we get out in relation to what we put in", or "How well do we or our agents convert inputs into outputs?" (Outputs are results delivered by the agents to an external party. This implies that efficiency examines the relationship between inputs and outputs; for example, planned versus actual delivery of milestones by service providers, or benchmarked comparison among programmes working to same or similar outcomes but using different pathways to achieve intended outcomes. To measure the efficiency of a partner in delivering the expected outputs, the outputs need to be clearly identified and quantified – and also need to be comparable.

- **Effectiveness**: Qualitative and quantitative measures of increase or decrease in outcomes that show that a programme is effective in delivering its intended objectives. This element examines the relationship between outputs and outcomes. How well do the outputs from an intervention achieve the desired outcome on poverty reduction for example?
- **Cost-effectiveness**: for example, how much impact on poverty reduction does an intervention achieve relative to the inputs that we or our agents invest in it?

Needless to say, these dimensions have to be considered together, not separately, balancing them to come to a judgment. This balancing act – for example reducing the costs of inputs and making efficiency savings can either support or undermine the VfM, why VfM does not need to be about monetizing everything and applying cost-benefit or cost-effectiveness analyses. This implies that effectiveness is not at odds with value for money, but rather an important component of it. For example, if the effectiveness of an activity is notably reduced because of a small cost saving, VfM is reduced. Similarly, while an activity may be very cheap and run efficiently, if it does not achieve results, it is not value for money. The quality of the outcomes is fundamental to understanding whether something is providing value, why economy is neither necessary nor sufficient to achieve higher-level efficiency or cost-effectiveness. Nonetheless, in a bigger perspective, effectiveness and cost-effectiveness are essentially what VfM is all about – since effective programmes are what is ultimately desired. Unfortunately though, this is much harder done than said. But there are several tools that can be used to assess effectiveness and cost-effectiveness in programming.

- **Step 4: Identifying evidence required:** In a logical and sequential process of evaluation design, it is only after clarifying the theory of change, criteria, and standards that relevant indicators and other sources of evidence can be identified. The preceding steps are important to help ensure that the evidence is relevant, measures the right changes, and is appropriately nuanced. This step involves determining what evidence is needed and will be credible to address the criteria and standards. This involves systematic analysis of the criteria and generally involves asking questions such as: How will we know? What would be credible evidence? What evidence is feasible to collect? (King et al., 2013). What counts as credible evidence to tell us whether we are seeing poor, adequate, good, or excellent VfM performance in the programme? This requires consideration of not just things that can be counted, or data we happen to have available to us, but what really matters and has some relevance to the VfM of the programme. Ideally this will include a mix of numeric, qualitative, and economic/monetary evidence. Key sources of evidence to support the VfM assessment will vary with context but may for example, include programme financial accounting data, log frame indicator data, programme operational data/reports, M&E data/reports, and results of economic analysis.
- **Step 5: Gathering evidence:** This involves following the same principles of good project management and fieldwork that would be followed for any evaluation or research project. Indeed, if the VfM framework is well aligned with wider M&E frameworks, the VfM assessment may be able to make use of relevant M&E data, minimizing duplication of effort. As with any evaluation, it is important to consider the counterfactual when making judgements about the attribution of results to a programme – i.e. what would have happened without the programme? In some programmes there will be experimental (e.g. randomized controlled trial) or quasi-experimental evidence to support causal inferences. In other cases, it will be necessary to apply a theory-based approach.

- **Steps 6–7: Analysis, synthesis, and judgements:** These stages involve three discrete, sequential steps: *i*) analysis of each stream of evidence (quantitative and qualitative) is first carried out to identify individual findings and themes; *ii*) synthesis is then undertaken to triangulate and consider the totality of evidence collected, including any areas of corroboration or contradiction between evidence sources. This is done firstly for each VfM criterion individually (i.e. economy, efficiency, effectiveness, cost-effectiveness and equity); and secondly for VfM overall; and *iii*) judgements are then made against the criteria and standards (i.e. the rubric) to determine and transparently report the VfM of the programme. Critically, the overall judgement of VfM is not required to be a straight average of the scores for each component; rather, greater weight should be given to those criteria that are deemed more relevant at the time the VfM assessment is carried out. For example, if good effectiveness and cost-effectiveness evidence is available, these criteria would generally be given greater weight than indicators focusing on inputs (e.g. economy) and outputs (e.g. efficiency), which receive more weight in the early stages of implementation.
- **Step 8: Reporting:** King argues that a good VfM report shall tell a compelling performance story, focused on and structured around the aspects of performance that matter (as defined by the criteria) and presenting a clear judgement about the level of performance (as defined by the standards). It should also give clear answers to important questions – by getting straight to the point, presenting transparent evidence, and being transparent about the basis upon which judgements are made (McKegg et al., 2017). Accordingly, VfM reports should be structured around the overarching VfM criteria (the Four E's), addressing each 'E' systematically in turn. Opportunities to improve the VfM process of making judgements about VfM against the criteria and standards provides a systematic way to identify areas where a programme is performing well and, potentially, areas where improvements can be made to strengthen VfM.

2.2. Proposed methodology for a rapid ex-post VfM assessment of CRP Livestock

2.2.1. Purpose of the evaluation

The primary purpose of this evaluation is summative. That is, we will conduct a rapid ex-post impact assessment of the various streams of research (flagships) carried under the CRP Livestock during the current cycle of the program to inform CRP Livestock regarding, from an independent perspective, whether and the extent to which the program has had the desired outcomes and effects on the beneficiaries (individuals or institutions). As discussed in the scoping review (Section 1), the VfM assessment framework is one of the applicable methods and designs for evaluating impacts of a programs such as the CRP livestock, and it has been used in impact assessment process across Rural Policy Research (RPR) (Slade et al., 2020), Food Fortification Programme (FFP) by DFID (Balagamwala et al., 2019) and Commercial Agriculture for Smallholder Farmers (CASH) project (Ghosh & Weatherhead, 2015). The evaluation dimensions of the VfM assessment framework include relevance, coverage, effectiveness, efficiency, impact, and sustainability. Much emphasis is placed on the impact dimension of the framework. The impact areas include observed economic impact, contribution to poverty alleviation, environmental impact, internationality of the problem and contribution to capacity building and research efficiency. To reflect more detailed flagship-level impacts, an inclusiveness criterion has been adopted, where impacts from each flagships stream are highlighted. The VfM assessment is disaggregated into research flagships. The five flagships include Livestock Genetics (FP1), Livestock Health (FP2), Feed & Forages (FP3), Livestock & Environment (FP4) and Livestock Livelihoods (FP5).

2.2.2. Conceptual framework

The evaluation will use a theory-based approach in that it uses CRP Livestock's ToC for each flagship as its conceptual framework to explain whether and how the ultimate outcomes have been achieved. The ToC will help systematically construct a plausible contribution story to draw causal conclusions about the how the programme achieved the intended ultimate outcome, and the different pathways used. To this end, we will draw on the programme's ToC to identify key issues the evaluation should address (i.e. evaluation questions); empirically verify outcomes and assumptions posited along the impact pathways in the ToC; and draw conclusions about whether, and how, the programme contributed to the observed results

2.2.3. Approach

The five flagships of the CRP Livestock, which consist of 21 product lines (see Table A in the appendix), will be the "unit of analysis" in this evaluation. Since many interventions carried out under these flagships do not readily lend themselves to monetarized or comparable effects (such those typically assessed using cost-benefit analysis or cost-effectiveness analysis), we will tailor an approach (based on the tools reviewed in the first phase of this study, including rapid appraisal methods, value for money, cost-benefit analysis, and outcome mapping) to provide a comprehensive quantitative/qualitative assessment of the impact of each flagship. Definitions of explicit criteria (aspects of performance) will be developed and aligned to ToC. Specifically, the evaluation of each flagship will be structured around six key evaluation criteria, which have been organized by six overarching OECD DAC evaluation criteria of relevance, coverage, effectiveness, efficiency, impact, and sustainability (Figure 5). These criteria will help provide an agreed and transparent basis for making judgements.

These criteria cover the five types of impacts identified in previous assessment of CRP Livestock, namely: (*a*) economic impact, (*b*) contribution to poverty alleviation, (*c*) environmental impact, (*d*) internationality of the problem and contribution to capacity building, and (*f*) inclusiveness related to youth and gender. Quantitative and qualitative data will be collected and compiled from programme implementation data, progress reports and other programme documents. Together criteria and dimensions of impact will provide an overarching framework for the evaluation, and form the basis of the evaluation design, data collection, and synthesis as outlined in Figure 4 below.

2.2.4. Evaluation questions and indicators

In accordance with the evaluation criteria, six *key* evaluation questions have been designed to evaluate each criterion at the flagship level. The key evaluation questions will then be further subdivided into *detailed evaluation questions*, which will be linked to the different parts and pathways of the ToC. For each flagship evaluation, these questions will be the basis for analyzing available qualitative and quantitative evaluation data to develop a finding against indicators. The evidence collected will be used to support the judgements made against each criterion (Figure 4). Table B in the appendix provides an example of evaluation criteria and questions based on FP1.

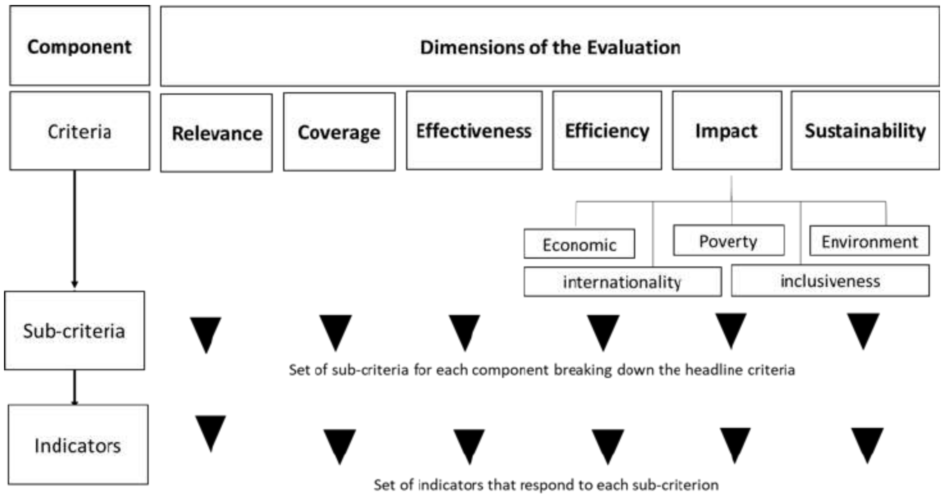


Figure 4. Evaluation criteria

3. Synthesizing Evidence: An ex-post VfM assessment of CRP Livestock

The purpose of this synthesis and deep-dive evaluation of the CRP Livestock is to assess the performance of the flagship programs and processes (accountability) and to learn what worked, where, for whom, and in what context (i.e. learning). As the scope of CRP Livestock flagship programs and its portfolio of product lines have grown, the study team has adapted an integrated VfM (as described in Section 2) and theory-based approaches to evaluating program effectiveness. These approaches allowed us to conduct a rapid ex-post impact assessment of the various streams of research (flagships) and product lines on the Strategy and Results Framework of the CGIAR focus areas including economic impact, poverty reduction, environment, internationality, and capacity development through CGIAR Impact areas such as creation of improved varieties, improved livelihoods and jobs, climate adaption and mitigation, environmental health and biodiversity, nutrition, health and food security, gender equality, youth and social inclusion (CGIAR-CO 2015a).

Unlike private sector investment projects, which are principally assessed against absolute economic and financial performance criteria and the extent to which they contribute to private sector development, this CRP Livestock program evaluation is assessed in relation to the program's relevance, coverage, effectiveness, efficiency, impact, and sustainability. Across the different program flagships, the assessment looks at whether and how the ultimate outcomes have been achieved using ToC. In this way, we systematically construct a plausible contribution story to draw causal conclusions about how the programme achieved the intended ultimate outcome, and the different pathways used. The evaluation approaches reflect and is harmonized with internationally accepted evaluation norms and principles, such as the quality standards for development evaluation of the OECD Development Assistance Committee, the good practice standards of the Evaluation Cooperation Group, and the norms and standards of the United Nations Evaluation Group. The IE team adheres to a multi-layered quality assurance model, which includes in-depth review of intermediate and final annual evaluation reports. Data sources include internal (CRP and CGIAR) data, external data sets, internal and external documentation. Evaluation design, including (e.g.) issues of scope, delimitation, data availability and analysis issues are discussed in this report.

3.1. Relevance of the CRP Livestock

CRP livestock covered several sustainable development target areas including poverty reduction, zero hunger, good health and well-being, gender equality, water and sanitation, responsible consumption and production, climate action and life on land. Nutrition-sensitive and cost-effective livestock and animal-source-food related interventions, including food technology, and policy interventions have widespread relevance to mixed dairy and small ruminant producers in rain fed and irrigated areas conducting mixed and livestock only farming. Malnutrition has significant implications for cognitive development of children, thereby contributing to poor educational outcomes, as well as implications for labor productivity of adults. This is largely a stream where value is captured through public health scoring in the environmental criteria. Improvements in human health are best captured using metrics such as disability adjusted life years.

An adoption ceiling was defined by specifying the extent of the problem being addressed by the CRP research streams (relevance domain) within each regional system and the maximum proportion of producers within that domain likely to adopt the end product or practice derived from each of the research streams. The extent of the problem within each production system was evaluated using benchmarks outlined in Thornton et al. (2000). They included the problem being described as widespread (83%), moderately present (67%), moderate (50%), limited–moderate (34%), limited (17%) or not present at all. Maximum adoption within each of the relevance domains is governed by demand factors, such as simplicity of the technology to be adopted, affordability and perceived need or usefulness by farmers. Supply factors include available infrastructure for delivery and presence of extension services. The CRP flagships consider gender relations, dynamics and norms that affect preferences, adoption and use of technologies, tools, products and services developed across different CRP streams. Dairy production in sub-Saharan Africa is the key domain for the research. A moderate (50%) proportion of mixed humid and arid systems (irrigated and rain fed), urban and other were relevant. A moderate (50%) of humid and arid mixed small ruminant systems in sub-Saharan Africa, Middle East and North Africa were estimated to be relevant. Regarding pig and poultry, half of urban, other, mixed arid and humid rain fed and irrigated poultry and a similar proportion of pigs in the same systems in sub-Saharan Africa were relevant domains. The research stream supports brooded and pre-vaccinated chicks that are adapted to typical low-input systems in poor rural communities. Similar pig genetic improvement was included, largely in Uganda. A low maximum adoption (5% of pig and poultry relevance domain) was assumed. Demand for improved animal health practices are widespread in mixed humid and arid (rain fed and irrigated) livestock systems in pig, dairy and small ruminant production across sub-Saharan Africa. A moderate (50%) relevance domain for ruminants and for pigs was assumed sub-Saharan Africa.

Peste des petits ruminants (PPR) can severely affect small ruminants in almost 70 countries in Africa, the Middle East and parts of Asia. A 34% relevance was included for East Coast fever (ECF), while widespread relevance of 67% was included for Africa and South Asia for PPR. Limited relevance of 17% was included in Asia for China. Contagious caprine pleuropneumonia (CBPP) afflicts cattle across a similar range. They mainly impact pastoral production systems. A moderate relevance of 50% was included for these systems. Live theileriosis vaccines have been commercialized but has not been widely adopted. Low adoption is thought to result from cold-chain requirements for live vaccine, substantial veterinary administration costs, potential for adverse animal reaction, strain specific immunity and possibility of introducing new ECF strains. PPR vaccines are used in Africa and Asia. CBPP vaccines are used in Africa. Low to medium adoption of 10% is assumed. Bovine and caprine mycoplasma diseases are widespread across Asia and Africa. ECF is found in 11 countries of eastern, central and southern Africa. Moderate-wide relevance is assumed for mycoplasma and African swine fever vaccines, while limited relevance is assumed for ECF across East and southern Africa. Vaccines are routinely used in Africa for CBPP. A vaccine does not yet exist for African swine fever. ECF usage is limited due to cold chain needs for the current live product. An adoption lag of 10 years is included. A long competitive lead time was included.

Brachiaria grasses in South and Central America have the potential to greatly increase carrying capacity in Latin America. The improved feeding strategies cluster was dominated by the nitrogen fixation in East Africa and a rangeland project was included in cluster 4 which was accommodated in this stream. Moderate relevance in livestock only arid and humid systems for nitrogen fixation in Africa, and livestock only arid ruminant systems for rangeland feed development are included as domains for this stream (50%). Some adoption among small ruminant producers in South Asia and Mekong was also included using cluster 4 funds. Emissions management research focused on livestock in smallholder systems and assess possible nutritional/management interventions which will reduce emissions. Intensive dairy and beef production systems in East Asia, Central America and Africa were the targets of this research program.

Improved water use investigated options for improving water and nutrient flows in priority sites. East Africa and Central America were a focus. Strategies were developed to ensure sustainable water use at basin and landscape scale, in terms of water quality and quantity. Half of mixed humid (rain fed and irrigated) systems in these regions could benefit from improved water flow management. Management practices also improve the nutrient flow and sustained watershed management in East African dairy. Regarding reduced greenhouse gas emissions; feeding and other farm practices to minimize greenhouse gas (GHG) emissions was implemented in the East African dairy field.

3.2. Coverage

This section presents the CRP Livestock coverage at the basis of each flagship, first at the regional level and then specifically in the priority countries based on various dimensions.

3.2.1. Regional level

The research developed under the Animal Genetics Flagship (FP1), adapted and extended tools to assess biological and socio-economic aspects of feed demand–supply scenarios, improve ration balancing, and enhance feed and forage substitution options. It extended the phenotyping capabilities for comprehensive feed, forage and fodder quality traits using stationary and mobile near infrared spectroscopy (NIRS) hubs in Africa, Asia and Latin and Central America, and improved cultivars increased feed and forage biomass availability in Latin America, South Asia, and Africa and to a limited extent in South East Asia. Improved 'full-purpose' nutritional quality for South Asia and Africa. Mixed humid and arid systems in Africa were the targeted for improved sheep breeds under FP1, as well as smallholder dairy in East Africa as the flagship also focused on African smallholder dairy productivity. Research and adoption lags of 3 and 10 years were included. Fifteen years were included before research benefit depreciation.

For the Livestock Health Flagship (FP2), context-specific herd health management packages were tested and adopted by farmers, extension and animal health workers in the following systems: Dairy in East Africa and India. Animal health/extension workers in selected priority locations used these new tools for identifying the most critical herd health interventions to be made. Hence, dairy in East Africa and India were the priority systems. Interventions were combined and tested in site-specific herd health packages. Furthermore, East Coast Fever or Theileriosis, primarily caused by the protozoan parasite Theileria parva, is found in 11 countries of Eastern, Central and Southern Africa — with around 30 million cattle being at risk from the disease. The disease is most important in small-scale intensive dairying and commercial cattle production systems where highly productive exotic breeds of cattle are raised.

For the Feeds and Forages Flagship (FP3), there were planted forage and 'full-purpose' crop development objectives in this stream. Brachiaria grasses was further developed for Central American conditions. Disease resistant Pennisetum was selected for East Africa and Opuntia (Cactus pear) cultivars for rangelands in South Asia. 'Full-purpose' crop development included millet and maize for South Asia and pigeon pea and beans for Africa and West Asia.

Regarding the Livestock Livelihoods and Agri-Food Systems Flagship (FP5), identifying priority interventions were likely to have a wide application across the target production systems. All systems covered in the CRP were assumed to be relevant, with 67% relevant domains for African and East Asian and Pacific systems. The regional focus of this flagship was in sub-Saharan Africa, South and Southeast Asia, and Central America. It worked to identify priority interventions that were likely to have wide applications across the target production systems. The regional focus of this cluster was in sub-Saharan Africa, Middle East and North Africa and Central Asia.

Lastly, animal health, forage and poor access to improved genetics are broad issues for priority production systems across Asia, Africa and Latin America. Half of mixed systems in humid rain fed and irrigated dairy in sub-Saharan Africa, 17% of mixed pig production in East Asia and the Pacific and half of sub-Saharan Africa; and small ruminant production in mixed and livestock only arid and humid systems of Sub-Saharan Africa and Middle East and North Africa were assumed to be relevant.

3.2.2. National level

The Animal Genetics Flagship (FP1) covered key target farming systems in Ethiopia, Tanzania, and Nigeria. In these countries, poultry and pig improvement, using mixed farming systems, were covered. Community-based breeding programs for small ruminants in low-input, communal grazing systems in Ethiopia were also piloted. Since 2009 Ethiopian community based sheep breeding programs are being implemented in three locations. System-wide adoption of this community programs were scaled up.

For the Livestock Health Flagship (FP2), context-specific herd health management packages were tested and adopted by farmers, extension and animal health workers for Ethiopian small ruminant and pigs in Uganda. Animal health/extension workers in selected priority locations used these new tools for identifying the most critical herd health interventions to be made. Small ruminant production in Ethiopia and pigs in Uganda were thereby the priority systems. Interventions were combined and tested in site-specific herd health packages.

Regarding the Feeds and Forages Flagship (FP3), the cluster covers livestock systems in priority locations in Ethiopia, Kenya, Tanzania, Burkina Faso, Nicaragua and Vietnam. Further analyses (with cluster 1 of the transformation and scaling flagship) informed work in other locations, especially in North and southern Africa, and Central Asia. This cluster developed technical solutions for rangeland systems of northern Kenya, Ethiopia, Burkina Faso, Nicaragua and Tunisia—enhancing longer-term productivity and reducing land degradation and biodiversity impacts. Half of mixed and livestock only arid (rain fed and irrigated) systems in these regions could benefit from improved rangeland management. Regarding reduced greenhouse gas emissions, enhanced water flow and improved natural resources management (NRM) in rangeland systems; feeding and other farm practices to minimize GHG emissions in Nicaragua cattle and Vietnamese pig production, along with improving longer-term productivity, was developed. Also, management practices improved nutrient flow and sustained watershed management in Nicaragua cattle and Vietnamese pig production, along with improving longer-term productivity. Finally, technical solutions embedded in management recommendations for rangeland systems of northern Kenya, Ethiopia, Burkina Faso, Nicaragua, and Tunisia were developed to enhance longer-term productivity and reduce land degradation and biodiversity impacts.

For the Livestock Livelihoods and Agri-Food Systems Flagship (FP5), Vietnam pig production, Ethiopia small ruminants, Kenya dairying and Uganda pig production were the key targets and priority interventions were likely to have wide applications across the target production systems.

3.2.3. Beneficiaries level

CRP Livestock implemented a range of initiatives, including facilitating community conversations, developing training modules and conducting training workshops (both in-person and through digital extension channels). CRP scientists and partners assessed capacity needs and conducted due diligence on solution providers in several countries across multiple livestock value chains. The CRP held short-term training and other activities with stakeholders including farmers, livestock entrepreneurs, extension officers and policymakers, reaching 13,938 individuals (34% women). During 2019, 32 undergraduate and graduate students also received long-term training, including 12 who completed their PhD studies (58% women). Furthermore, the CRP conducted due diligence on six stakeholders applying digital technology solutions to improve small holder pig farming and facilitated needs assessments from the flagships for various capacity development activities; for example, the “eWeigh” mobile application (app), which uses heart girth as a proxy measure to estimate live weight, was tested by Kenyan dairy farmers. Using community conversations, training modules for specific animal health issue areas were developed and rolled out, with new modules on antimicrobial use and animal welfare developed and tested. In Ethiopia for example, a module on community-based breeding programs was integrated into the livestock genetics curriculum of 3 universities and a tailored MSc training on breeding and genetics was introduced in 2 universities. This, in combination with improved scaling approach, led to high uptake of improved sheep fattening practices and technologies in Ethiopia. Multiple extension materials were also developed for forage training in Central America, Colombia and Tunisia, including e-learning training modules.

In 2020, the CRP reached a total of 30,930 individuals (33% women) during 2020 (Annual Report, 2020). Of these, 23,128 people (35% women) were engaged in long-term training, the majority (23,100) through a 4-month weekly virtual seminar series on Sustainable Beef and Dairy in Colombia, and the remainder 1 intern and 27 students in formal education (of whom 6 women and 4 men completed PhDs). Participants in short-term training and other events were 7,802 (30% women). Some key achievements in 2020 were the development of training course materials and guidelines for dairy cattle breeding in East Africa, training of Ugandan herd health champions at SLU, hybrid (remote/on site) training in prudent and medically rational use of antibiotics, training to improve the knowledge, attitudes, and practices of women regarding hygienic milk production and handling in Ethiopia.

Towards the first target, 100 million more farm households adopted improved varieties, breeds, trees, and/or improved management practices. CIAT’s existing Urochloa hybrids were scaled on approximately 100,000 additional hectares in 2020, reaching a total of at least 1,100,000 hectares in 30 countries. Over 140,000 farmers in East Africa are reported using digital tools for recording dairy performance of dairy cows and accessing knowledge services, supporting selection with genomic tools of appropriately adapted animals for breeding and establishing viable local artificial insemination services.

3.2.4. Gender dimension

Gender dimensions of livestock research was given more profound consideration in order to enhance equitable opportunities for both female and male participants in livestock communities. This implies that decision and policymakers raised the overall awareness of gender dimensions and also improved the institutional frameworks that support gender and social equity. Thereby, livestock-keeping communities now support and adopt gender equitable norms to a greater degree. Furthermore, gender analysis was integrated into tools and methodologies for priority setting, learning and assessing impact by developing gender sensitive indicators, and methodologies to assess progress on gender strategic change. More equitable asset allocation generates indirect productivity benefits in target systems.

Gender inclusion was considered by CRP-livestock from the proposal stage, gender analysis to improve technology development, and gender analysis to enhance progress towards gender equity were focual in the CRP program. The CRP considered specific constraints that women and men face to ensure the development of interventions that support gender equality and meet poverty reduction and productivity objectives. The CRP also focused on gender transformative approaches which assess how women are integrated into agricultural development. Specific research on gender and social equity was a cluster within the Livestock Livelihoods and Agri-food Systems flagship, while gender analysis has been integrated across all the flagships to varying degrees.

The Livestock Livelihoods and Agri-food Systems flagship had the highest overall score for inclusion. The gender cluster sits within the second of these flagships and many on-farm and scale up activities are pulled together amongst them. The Animal Genetics flagship considered gender relations, dynamics and norms that affect species and breeds preferences that in turn affect the relevance of genetic improvement outputs and their adoption.

The Livestock and Fish gender strategy noted that gender disparities also are evident in the types of livestock women and men own, with women more likely to own small livestock than large livestock. This flagship assessed species and breed preferences in relation to gender norms, and factor in gender dimensions to genetic improvement approaches and delivery mechanisms. Gender dimensions in animal health—such as preferences for vaccines and their utilization, impact of zoonosis on household health or access to veterinary services—was considered in the Animal Health flagship. Feeds and Forages considered gender dynamics in the choice of forage systems and crop and forage species preferences. These dimensions was factored into assessments that inform feeding options. Research into the roles of women and youth on the environment was integrated across the environmental flagship.

Research within the Livestock Health Flagship showed gendered decision making on the use of vaccines. Vaccines often had the consequence of increasing productivity for the household which resulted in increased labour for women, but with limited returns for them as their husbands are usually in control of the money resulting from the sale of animals. The consequences were that in some cases women were wary of vaccines. Research under the Livestock Genetics Flagship also explored the gendered trait preferences of chickens based on local use.

The growing body of work on gender across the flagships was further supported by the arrival of two post-doctoral fellows funded in part by the System Management Office. Both post-docs are part of the cohort on breeding: one initiated research on the gendered trait preference for small ruminants in the dry zones of Kenya, and the other began looking at trait preferences for men and women on forages in Kenya and Ethiopia.

Furthermore, a study in Afghanistan identified constraints and opportunities for women's involvement in the fodder production systems (Annual Report, 2017). The Women's Empowerment in Livestock Index (WELI) was tested. WELI was next implemented in Honduras to understand the variation within different systems. Additionally, the use of the WELI in 4 projects has resulted in a larger WELI database and the ability to compare data across 6 countries (Nepal, Senegal, Uganda, Ethiopia, Kenya, Ghana and Tanzania). These results are important in understanding how livestock can be transformative within a gender context.

In 2017, CRP explored linkages between gender, livestock and nutrition where research explored the methodological implications of the connection between women's empowerment and household food and nutrition security through qualitative and quantitative approaches (Annual Report, 2017). Research was also initiated on gender at landscape level through engagement of the gender team in the development of national livestock master plans. To promote continued capacity development in mainstreaming gender, the gender team was embedded within the different flagships providing individual coaching to specific scientists. Several training workshops were undertaken on gender and livestock: two in collaboration with the FAO at a workshop in Colombo for government livestock experts in the region, which was so popular the Sri Lankan government requested a follow-up training for its livestock department. Some 40 staff members from ILRI's Accelerated value chain development project in Kenya were also trained on gender and livestock.

In 2018, the Livestock and Environment Flagship expanded work on rangeland management to include more gender dimensions in response to partner interest expressed through the International Land Coalition Rangelands Initiative (Annual Report, 2018). Livestock CRP incorporated gender dimensions in rangeland management tools.

The CRP Livestock also hosted the 2018 CGIAR gender platform scientific conference at ILRI in Addis Ababa. Furthermore, gender transformative work was done in Ethiopia using community conversations to understand how men and women deal with animal diseases. This allowed them to understand each other's contributions and has resulted in a more equitable sharing of roles. The continued work with the Ethiopian Agricultural Research Institute on gender has also meant that the institute has increased the gender balance of its researchers, and also has become more aware of how women fit within the agricultural sector.

Gender research within the CRP Livestock was able to thrive in 2019. It continued to grow both in breadth and in depth. Gender work in the flagships was consolidated through the allocation of dedicated gender researchers: 30% staff time in the Livestock Health Flagship, 50% staff time in Feeds and Forages Flagship and a full-time staff member in the Livestock and the Environment Flagship.

3.2.5. Youth strategy

In line with the current framing in popular discourse of youth as an employment issue, the CRP's focus on young people has been formulated to revolve around "employment, entrepreneurship and capacity development". In 2017, the CRP developed a strategy for addressing the issue led by a youth specialist as a joint appointment with Royal Tropical Institute (KIT). Primary qualitative data was collected in four CRP priority countries (Tanzania, Uganda, Ethiopia and Nicaragua) and a framework paper drafted to provide a coherent analysis of the emergence of youth on the development agenda, a conceptualization of youth and an analytical framework to guide future work on youth within the Livestock CRP (Annual Report, 2017). Three pathways developed to increase youth engagement in CRP streams include:

- Improved adoption of technologies and management practices through youth-mainstreaming and youth-responsive design
- Strategic research to increase knowledge on youth and livestock, especially with respect to land tenure and migration
- Contributing to youth employment, which should be a focus of the Livestock CRP.

In 2018, the CRP held an e-consultation on youth to validate its emerging youth strategy (Annual Report, 2018). Livestock CRP research included work that focused on various 'left behind' populations, which include youth, small and middle scale livestock keepers and other actors, and communities in pastoral areas. The Feeds and Forages Flagship established a new partnership with one of the largest dairy companies in Colombia to support their youth-centered education program of dairy producers with specific technical and social components in 2020. In Ethiopia, ICARDA in partnership with the National Research Centers and the national extension trained youth as disseminators of improved sheep fattening technologies and practices in the Ethiopian highlands: 485 youth in 75 villages were organized into 44 youth groups to serve as benchmark sites for sheep farming communities to adopt improved sheep fattening practices and undertake sheep fattening as a business. To ensure sustainability of the entrepreneurial culture, community of practice teams in target sites were formed and strengthened with knowledge and skills to enable them to garner support from regional and local administrative offices to create an enabling environment for commercialized sheep fattening. The Livestock and the Environment Flagship, working with CCAFS, conducted research on how to more effectively include youth in dairy value chains, paying particular attention to how interventions to make these value chains more "low emissions" might affect youth. Based on the CRP youth strategy, the Livestock Livelihoods and Agri-Food Systems embarked on compiling existing data and information from projects.

In 2020, seven young agripreneurs in Tunisia received CRP-funded grinders to develop their feed businesses. They were trained on technical issues like feed composition and two of them began selling their own produced feed comprising mixed ground maize, barley and faba beans. Furthermore, community conversations activities took place in Ethiopia, wherein youth were specifically targeted. The CRP was also working on the participation of youth and ethnic minorities in livestock activities in Vietnam.

The Sustainable Landscape project in Nicaragua continued a strong emphasis on the participation of women and youth, and promotion of women's entrepreneurship. Of the farmers participating in Farmer Field Schools, 52% were women. An in-depth analysis of the youth-centered education program for dairy producers in Colombia found that:

- A quota system implemented for training events reduced barriers to women’s participation
- Addressing key areas of disempowerment, identified in consultation with 14 female livestock keepers, enhanced women’ decision making power in households and communities

3.3. Effectiveness

3.3.1. Target production systems and relevance domain

The global livestock production system was disaggregated into the agro-climatic classification by Sere and Steinfeld (1996) for World Bank regions and typical benefits estimated for producers within each respective system. Livestock product volumes were estimated for each system using FAOStat data and ILRI spatial databases for 2012 (FAO 2016). An adoption ceiling is defined by specifying the extent of the problem being addressed by the research stream within each regional system and the maximum proportion of producers within that domain likely to adopt the end product or practice derived from each of the research streams. The extent of the problem within each production system was evaluated using benchmarks outlined in Thornton et al. (2000). Maximum adoption within each of the relevance domains is governed by demand factors, such as simplicity of the technology to be adopted, affordability and perceived need or usefulness by farmers. Supply factors include available infrastructure for delivery and presence of extension services. These benchmarks are again used in this impact assessment.

3.3.2. Productivity gains

Gains were estimated for an average animal in each system adopting the technology at scale. They were typically lower than those observed during on-station trials, as other constraints such as nutrition, herd health and stresses hinder the potential gains a technology may provide. Yield gaps have been estimated for Indian and Ethiopian livestock producers as part of the recent Bill & Melinda Gates Foundation supported Live GAPS project. For example, improved feeding and promoting changes in the herd structure towards more crossbreeding was estimated to increase total Indian milk production by 60% in 15 years (Herrero et al. 2015). More conservative estimates of productivity gain are included in this impact assessment. They range from a very high productivity gain assumption of 15–20% for genetic improvement of crossbred poultry in Africa and planted forages in Central America to indirect productivity gains of 0.1% where research is indirectly impacting productivity through priority setting.

Benchmarks from Thornton et al. (2000) are generally used in this impact assessment. They include a high productivity gain of 5% in a small target area, medium gain of 3% in well-defined target areas, low as a 1% direct productivity gains and low–low (0.1 %) indirect productivity gains in a large area where impacts are difficult to define. These productivity gains are diluted by adoption and probability of success; so expected gains are far less than those of the LiveGAPS project. Estimated portfolio returns in the impact assessment are, however, in line with plausible estimates of returns for CGIAR research outlined by Raitzer and Kelley (2008). These authors reviewed impacts from the suite of completed CGIAR economic impact studies until 2008. Comparison of their results with those in this impact assessment is discussed in the results section.

Improved breeding program efficiency and targeting through the use of livestock system characterization and intervention analysis, enhanced repository of information on distribution, and assessment of genome diversity. The stream improved the productivity gains in the poultry, dairy and small ruminant research undertaken than would otherwise have been the case. Productivity improved by an additional 1%. Furthermore, as dairy production in sub-Saharan Africa is the key domain for the research, a moderate (50%) proportion of mixed humid and arid systems (irrigated and rain fed), urban and other were assumed to be relevant. The productivity gain was high, (20%).

In Ethiopia early research found live weights (three and six months' weights) were heavier and gains (pre- and post-weaning weights) were faster (at least $p<0.05$) for the base year of the community trial in that country. Results have improved as the trial has progressed. A 40% increase in overall off-take was targeted for health, nutrition and breeding improvement. A 15% improvement is assumed for genetic improvement. Since 2009, Ethiopian community based sheep breeding programs are being implemented in three locations. A low-maximum adoption target (5%) of the relevance domain is estimated. Research and adoption lags of 2 and 20 years are assumed. Adoption lag was long due to the time to expand participating communities.

The Livestock Health Flagship (FP2) Research involved risk assessments and studies on risks of emerging diseases, including tick distribution and vector-borne diseases. All systems being targeted by research in clusters 2–4 benefit from improved strategy and disease management priority setting. It was estimated that the productivity benefits of research in these clusters was improved by 0.1% as a result of the research.

Regarding the Feeds and Forages Flasgship (FP3), there were planted forage and 'full-purpose' crop development in this stream. Brachiaria grasses was further developed for Central American conditions. Disease resistant Pennisetum was selected for East Africa and Opuntia (Cactus pear) cultivars for rangelands in South Asia. 'Full-purpose' crop development includes millet and maize for South Asia and pigeon pea and beans for Africa and West Asia. 50% of mixed humid systems in the Americas are relevant for planted forages and a similar portion of mixed systems in South Asia and North Africa for full-purpose crops. Brachiaria grasses in South and Central America have the potential to greatly increase carrying capacity in Latin America. Productivity increases of as high as 300% were stated by the research team. Productivity increases of 20% are included for this species and 3% productivity increases are included for the other planted forage and full-purpose crop development options. A maximum adoption of 5% of relevant domains was included. Research and adoption lags of eight and 10 years were included.

Furthermore, the improved feeding strategies cluster was dominated by the nitrogen fixation in East Africa and a rangeland project is included in cluster 4 which is accommodated in this stream. Moderate relevance in livestock only arid and humid systems for nitrogen fixation in Africa, and livestock only arid ruminant systems for rangeland feed development are included as domains for this stream (50%). Some adoption among small ruminant producers in South Asia and Mekong is also included using cluster 4 funds, resulting in a 3% productivity gain.

For the Livestock Livelihoods and Agri-Food Systems Flagship (FP5), animal health, forage and poor access to improved genetics were broad issues for priority production systems across Asia, Africa and Latin America. Half of mixed systems in humid rain fed and irrigated dairy in sub-Saharan Africa, 17% of mixed pig production in East Asia and the Pacific and half of sub-Saharan Africa; and small ruminant production in mixed and livestock only arid and humid systems of Sub-Saharan Africa and Middle East and North Africa. Technologies, management strategies and institutional arrangements developed through livestock systems optimization was tested in priority locations. Farm productivity was estimated have increased by 3% through improved crop tree livestock interactions in mixed systems and herd and grazing management in pastoral systems.

Improved management options and products were adopted by a limited proportion of the sector. Community-based breeding, new diagnostics, animal health packages, forages and enhanced management practices adoption was limited by the capacity of extension to diffuse results. The maximum adoption was estimated to be 5% across relevance domains of all target systems. Research and adoption lags of five and 10 years were included.

3.4. Efficiency

Some relevant efficiency gains were achieved by CRP in 2019. The first relates to the joint development of greenhouse gas assessment methodology and capacity by both CCAFS and the Livestock CRP, reducing duplication of effort and cost (Annual Report, 2019). Second, investment in IT solutions for information collection, management and application for detection and response to implementation problems in large scale projects is reducing replicated efforts and improving effectiveness. Examples include use of cloud computing to reduce IT administration overheads, use of database-driven dashboards to summarize project status and the creation of opportunities for data sharing and interoperability with partner IT systems. Such gains have been achieved by the Livestock Genetics flagship with the National Animal Genetic Improvement Institute in Ethiopia and Green Dreams in Kenya and Ethiopia. The International Livestock Research Institute (ILRI) and CIAT, which partner in the Feeds and Forages Flagship, brought together their complimentary capabilities in the ILRI forage genomics and the CIAT forage breeding programs to support each other in areas including plant genotyping and forage breeding/pre-breeding and thereby avoid duplicating investment. Both centres also contributed to the development of the Tropical Forage Germplasm Strategy in consultation with the Genebank Platform. By prioritizing forages likely to be in high demand, the link between germplasm conservation and its use in the Livestock CRP has been strengthened.

The Feeds and Forages Flagship sought strong interaction and collaboration with the CCAFS CRP, for example on the development of forage materials for regions strongly affected by climate change and research on the Biological Nitrification Inhibition phenomenon in grasses and soil-grass interactions. The Breeding Program Assessment Tool (BPAT) developed by the University of Queensland in Australia was applied at CIAT in 2018 identifying areas for improvement in the Centre's forage breeding programs. This resulted in a restructuring and improvement process from 2019 onwards and led to significant efficiency gains (Annual Report, 2018).

In Livestock Livelihoods and Agri-Food Systems Flagship, time and effort efficiency gains were achieved by using common sampling procedures for some research projects. Expanded processing modules for Rural Household Multi-Indicator Survey (RHoMIS) data, especially for transforming data sets between database formats, have improved data cleaning and extracting steps. Quantitative impact assessments (of smart marketing and value chain interventions) in Ethiopia were carried out through direct data recording, which required more time in survey preparation in 2018 but greatly improved speed and quality control during survey implementation (Annual Report, 2018). To strengthen the time commitment of flagship leaders and administrators to planning, reporting and managing flagship teams, flagship management costs were increased to support a larger share of their time (30%).

In 2020, strong synergies were sought between the Feeds and Forages and Livestock and the Environment Flagship, and CCAFS to jointly work on greenhouse gas emission assessments, youth integration and gender (Annual Report, 2020). In the Livestock Genetics Flagship, increased investment was made in information technology (IT) solutions to reduce replicated efforts and allow a more agile detection and response to problems in implementation of large scale bilateral projects.

Examples include: improved database architecture; use of cloud computing to remove the IT administration overhead; use of database driven dashboards to summarize project status; and integration of the IT systems of partners, notably National Animal Genetic Improvement Institute in Ethiopia and Green Dreams in Kenya and Ethiopia.

3.5. Impact

3.5.1. Impact of CRP Livestock Genetics Flagship (FP1)

Livestock Genetics (FP1) focused on implementation of genetic improvement programs and associated delivery systems to ensure that smallholder farmers utilize appropriate livestock breeds. In line with the impact dimension of the VfM framework, the types of impacts identified from existing CRP Livestock annual reports and databases include (a) contribution to poverty reduction, (b) contribution to hunger and food insecurity reduction, (c) income, (d) education, knowledge and innovation, (e) capacity development. Figure 5 presents a systematically constructed and a plausible contribution story to show the how the Livestock Genetics Flagship achieved the intended ultimate outcome and impact. As shown in Figure 1, the main output from FP1 was access to improved animal genomic and genetic resources. These genetic improvement strategies for improved livestock genetics implemented by national research and development partners, and the private sector in six CRP priority countries and other locations.

In the short and intermediate terms, the impact of access to access to improved animal genomic and genetic resources includes:

- **Closed yield gaps through improved agronomic and animal husbandry practices**: Superior breeds with improved yields were developed and disseminated. For instance, Tanzania's national artificial insemination centre adopted genetically superior crossbreed dairy cattle. There were upscaling of community based breeding programs (CBBPs) in several African countries. Private poultry companies in three countries adopted more productive and resilient chicken breeds for delivery to smallholder farmers. FP1 also established a network of reproductive laboratories for small ruminants in Ethiopia and Tanzania, and the establishment of smallholder poultry platforms in Ethiopia, Nigeria and Tanzania.
- **Development of digital tool for recording performance:** FP1 developed a mobile application for capturing dairy-cattle performance and providing feedback is in use by 140,000 East African livestock keepers. Towards genetic improvement strategies, a genomic tool for determining the breed composition of dairy-cattle was developed and the selection index for the Tanzanian dairy-cattle breeding program updated.
- **Increased conservation and use of genetic resources:** In relation to poultry genetic improvement, research into more productive, resilient and farmer-preferred chicken breeds (a flagship innovation) has resulted in delivery of improved birds to Ethiopian, Tanzanian and Nigerian smallholder farmers by private companies.
- **Increased livelihood opportunities:** The livelihood opportunities for poor livestock keepers increased through the different product lines and activities.
- **Access to technologies that reduce women's labor and energy expenditure:** Women resource poor livestock keepers sustainably utilized improved livestock genetics, both productive and adapted, in 3 priority countries. (F1 Outcome: 1.4 -2019 Annual Report, pg 28).

The long term impacts of access to access to improved animal genomic and genetic resources include:

- **Poverty Reduction:** In the CRP priority countries, FP1 contributes to reduction in poverty among livestock farmers through the closure of yield gaps through improved agronomic and animal husbandry practices, development and use of digital tools for recording performance, increased livelihood opportunities.

- **Hunger/food insecurity reduction:** In terms of hunger and food security, this flagship contributes to its reduction through closure of yield gaps through improved agronomic and animal husbandry practices, increased livelihood opportunities and increased conservation and use of genetic resources. Girard et al. (2018) found that women's empowerment is associated with maternal and child diet diversity in pastoral communities in Tanzania. A survey by Galiè et al. (2019) on this stream examined the relationship between women's empowerment, household food security, and maternal and child diet diversity (as indicators of nutrition security) in two regions of Tanzania. Using the household food insecurity access scale, Galiè et al. (2019) found that all three empowerment domains were positively associated with food security and nutrition in the qualitative analysis.
- **Income:** The animal genetics flagship contributes to income generation in the long term. As shown in Flagship #1 (Figure 5) Theory of Change, the income gains are achieved through closing of yield gaps through improved agronomic and animal husbandry practices, improvement in performance through the use of newly developed digital tools, increased livelihood opportunities and through the use of technologies that reduce women's labour and energy expenditure. The extent and magnitude of increase in income will require further quantitative impact assessment.

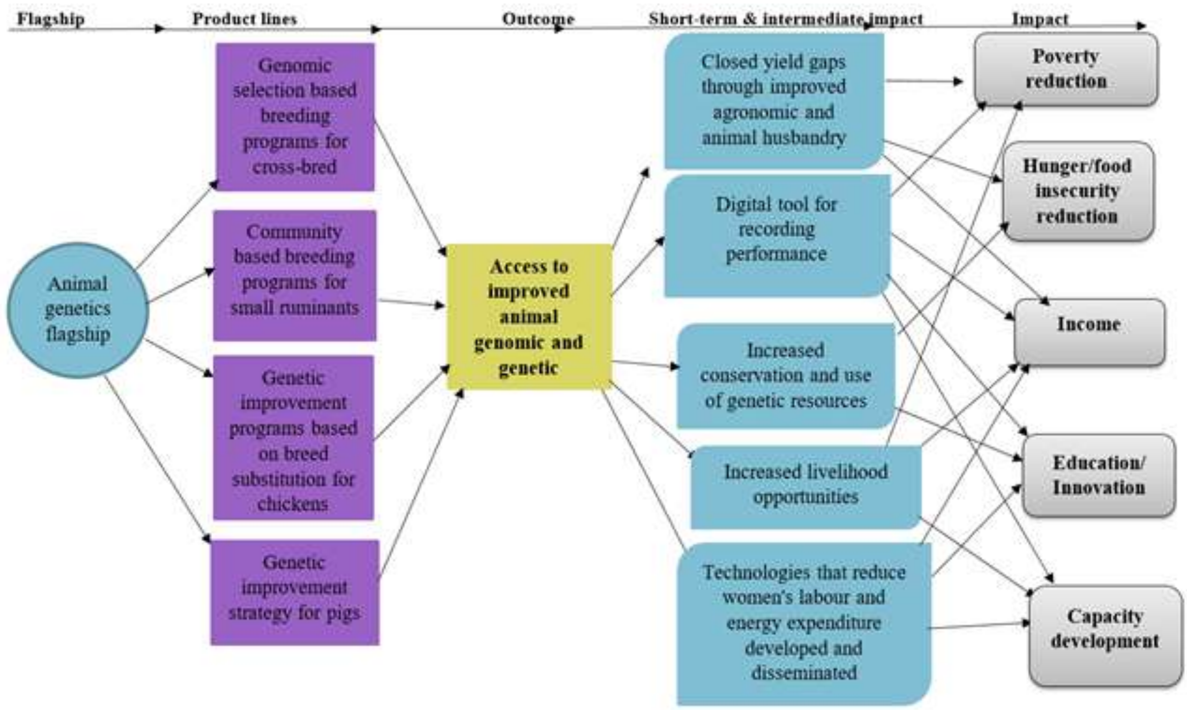


Figure 5. Detailed short-to-medium term and final impacts of Animal Genetics flagship

- **Education/innovation:** The development of digital tools for recording performance, genetic resources and technologies for reduce women's labour and energy expenditure contributed to education and innovation. For instance, the flagship developed Community Based Breeding Programs (CBBPs) for small ruminants being upscaled in a number of African countries (OICR 3269). To contribute to educational development, the CBBPs were incorporated in the curriculum of national universities (OICR 3271). To facilitate upscaling, educational training were offered to the CBBPs (2020 Annual Report). The contributing training material on CBBPs were incorporated in the curricula of Ethiopian national universities in Ethiopia. To deliver improved genetics in emerging dairy systems, a viable and sustainable business model based on fixed-time artificial insemination technology through a mix of public-private actors was developed and tested in Kenya. In addition, a digital tool for interaction with farmers, and a protocol and pipeline for use of genomic information in the breeding program were developed (2019 Annual Report FP1 - Livestock Genetics). Over 140,000 farmers in East Africa are reported using digital tools for recording dairy performance of dairy cows and accessing knowledge services, supporting selection with genomic tools of appropriately adapted animals for breeding and establishing viable local artificial insemination services.

- **Capacity development:** Through the development and establishment of digital tools, technologies and laboratories, the capacity of partner universities and researchers were improved. For example, nine reproductive laboratories were established to support dissemination of improved small ruminant genetics from breeding programs in Ethiopia and Tanzania (1.3.3). Thus, the flagship has contributed to enhancing the institutional capacity of partner research organizations, enhanced individual capacity in partner research organizations through training and exchange, increased capacity for innovations in partner research organizations and increased capacity for innovation in partner development organizations and in poor and vulnerable communities.

3.5.2. Impact of Livestock Health Flagship (FP2)

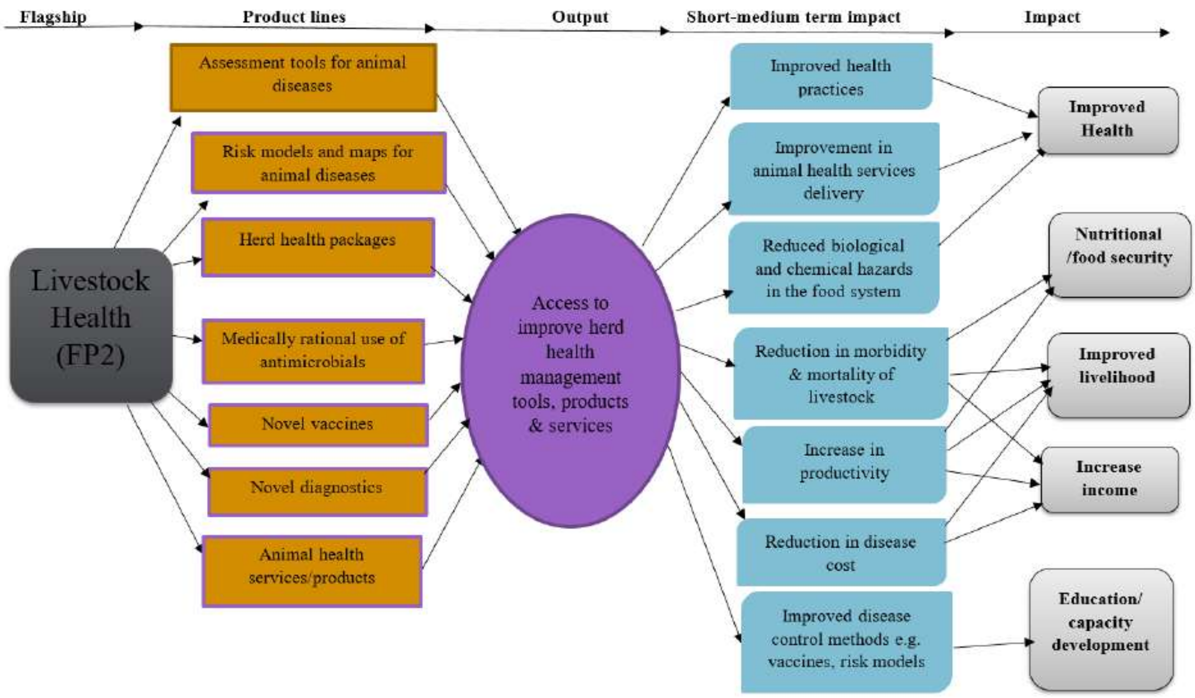
The main output, short-to-intermediate and long term impacts realized from Livestock Health (FP2) are presented in Figure 6. The main output from this stream was access to improved herd health management tools, products & services. This output was collectively attained through different activities including (i) Development of tools for assessing socio-economic impact of diseases (ii) Development of disease risk models and maps (iii) Development and evaluation of herd health packages (iv) Establishment of protocols for survey of antimicrobial use and for monitoring AMR (v) Development of improved vaccines against East Coast fever (ECF) (vi) Development of improved diagnostics and diagnostic platforms (vii) Development of improved vaccines against different diseases (viii) Animal health service and products provision.

The short-term and intermediate impact of FP2 include the following:

- **Improved health practices:** The livestock health stream has contributed to the use of improved health practices that reduced women's labor and energy expenditure by approximately 10% in across the countries where this stream was implemented. It is worth mention that the reduction in women’s labor and energy expenditure will differ across countries and product lines. About 2.8 million women across 11 countries are anticipated to be reached (Livestock-Sub-IDOTargetsAH_19Mar16+Data). The improved use of health practices is as a result of availability of tools, models and maps for disease detection and for assessing socio-economic impact of diseases as well as protocols for survey of antimicrobial use and for monitoring AMR.
- **Improvement in animal health services delivery:** The delivery of health services to farmers have improved considerably because of improved access to animal health services and products, health packages, models and maps for disease detection and availability of newly developed vaccines against lethal East Coast fever in cattle, Contagious Caprine Pleuro-Pneumonia (CCPP) and African Swine Fever (ASF).
- **Reduced biological and chemical hazards in the food system:** The use of protocols for survey of antimicrobial use and for monitoring AMR, medically rational use of antimicrobials and advanced development of a safe and effective Rift Valley Fever vaccine for livestock translates into reduction in harmful substances in food. Thus, this leads to improvement in food safety

in over 5.6 million people in livestock keeping households and for 2.7 million consumers in 7 countries (Livestock-Sub-IDOTargetsAH_19Mar16+Data).

- **Reduction in morbidity & mortality of livestock:** The product lines in this stream contributes



to reduction in reduction in morbidity and mortality of livestock and reduction in disease control costs through early diagnosis of disease, impacting 6.4 million people in livestock keeping households, across 10 countries.

- **Increase in productivity:** The productivity of about 1.6 million livestock keeping households (4 million individuals) are increased through the use of integrated herd health packages in 9 countries, use of improved vaccines, improved disease diagnostics and better animal health service and products provision.
- **Reduction in disease cost:** The livestock keeping households in the priority countries for this stream experienced reduction in prevalence of zoonotic pathogens and other livestock diseases. The reduction in prevalence of diseases translates into reduced veterinary, medication and treatment cost. The application of rational use of antibiotics in the livestock food system also, translates into reduced risk for increase in anti-microbial resistance. Case study: A case study in India using 954 dairy animals in Bihar and 1348 dairy animals calculated the sum of income losses and expenditures incurred. The major cost incurred resulted from extended calving intervals (46.1% of the total cost), followed by loss through salvage selling (38.1%), expenditure for treatment of repeat breeders (5.9%), loss of milk production (5.3%) and expenditure for extra inseminations (2.0%). The estimated cost of reproductive problems was Indian Rupees (INR) 2424.9 (USD 36.1) per dairy animal per year (of the total dairy animal population) which represented approximately 4.1% of the mean value loss of dairy animals (INR 58,966/USD 877) per year (Deka et al., 2021). With this stream, livestock farmers who benefited from these stream are able to experience a reduction in these cost.

Figure 6. Detailed short-to-medium term and final impacts of Livestock Health flagship

- **Improved disease control methods:** Livestock keeping households in the priority countries have been exposed to improved disease control methods through activities such as the establishment of protocols for survey of antimicrobial use and for monitoring AMR, development and use of improved vaccines, risk models and maps for diseases.

In the long term, the impacts realized from Livestock Health (FP2) are categorized into:

- **Improved health:** In the long term, improvement in health practices and animal health services and product deliveries contributes to improvement in livestock health. In addition, reduction in biological and chemical hazards in the food system translates into improvement in food quality and safety for consumers. Herd health guides compiled by the Livestock Health Flagship for priority countries production systems demonstrates the better balance achieved between individual disease versus integrated health management, including notably more efficient antimicrobial use. A survey toolbox was also finalized to support the global effort to eradicate Peste des petits ruminants (PPR).
- **Nutritional/food security:** Reduction in morbidity & mortality of livestock and increase in productivity contribute to increase in livestock products such as milk, eggs, meat, etc. Thereby enhancing food and nutritional security in priority countries.
- **Increase income:** The income of livestock keeping households in the priority countries are increased through reduction in morbidity & mortality of livestock, increase in productivity and reduction in disease cost. It is noted that many of the poor in these areas are predominately smallholder farmers, and that increased productivity may be an important pathway to poverty reduction if it leads to increased incomes.
- **Improved livelihood:** Increase in total household income from livestock-related activities, including increase in proportion controlled by women, for households & individuals in the priority countries. Livestock keeping households realized increase in income of the household enterprise from chicken, pigs, small ruminant and dairy cattle, respectively, through the use of health livestock herds combined with other appropriate animal husbandry practices, across priority countries.
- **Education/capacity development:** This stream has improved education and knowledge on livestock disease detection, diagnosis, and mapping and vaccine development. Assessment tools for significance of animal diseases and risk maps for emergence of animal diseases are used by 100 local and national and 50 international research partners and donors for priorities research and development interventions to reduce livestock disease risks for livestock keepers. For example herd health in small ruminant production was promoted in Ethiopia through community-based gastro-intestinal parasite control programs, and community conversations–a novel participatory farmer training approach. A data collection tool on knowledge, attitudes and practices on farmers' use of antimicrobials was developed and applied in three countries: With CRP A4NH, a CGIARwide strategy for Anti-Microbial Resistance (AMR) was developed in India. In addition, a model was established for testing Contagious Caprine Pleuro-Pneumonia (CCPP) vaccines, a possible vaccine target for Contagious Bovine Pleuro-Pneumonia (CBPP) and proof-of-concept for an East Coast fever (ECF) vaccine was developed. Two novel delivery models of animal health services and products and cap dev/training methods tested in collaboration with partners in Kenya, Tanzania, Ethiopia and Mali in 2018.

3.5.3. Impact of Feeds and Forages Flagship (FP3)

The impacts of feeds and forages (FP3) stream are summarized into short-medium and long term impacts in Figure 7. In the short-term and intermediate impacts, the impacts of FP3 mainly include the following:

- **More efficient use of inputs:** Smallholder livestock keeping households (individuals) efficiently using inputs through optimized feeding strategies, including rations and processing in priority countries. About 1.13 million poor households (5.1 million individuals) efficiently using inputs through optimized feeding strategies, including rations and processing across 11 countries. In 2018, Urochloa hybrids developed by Feeds and Forages FP3 were scaled up on approximately 100,000 additional hectares in 15 countries. The total area sown with CIAT hybrids under FP3 Feeds and Forages was estimated at 1 million hectares in 51 countries by the end of 2019. Local, national and international research and development partners, the private sector, decision-makers and livestock producers are able to diagnose feed constraints and opportunities and to effectively prioritize and target feed and forage interventions, resulting in: a 10% improvement in utilization of feeds and forages.
- **Closed yield gaps through improved feeding options and strategies:** In terms of yield, about 1.18 million poor households (5.4 million individuals) realized 30% increase, on average, in productivity through the use of improved feeding options and strategies, in 12 countries. To promote uptake of dual-purpose crop cultivars (3.4), dual-purpose maize hybrids were disseminated on 100,000 hectares (India). To promote improved forages (3.3), global private sector commercialization of existing Urochloa hybrids was increased, new generations of Megathyrsus/Urochloa targeting specific production niches in Latin America/Caribbean and East Africa (LAC/EA) developed, 3 new barley hybrids released (India/Jordan), and candidate genes associated with Napier forage traits identified.
- **Reduced market barriers:** Livestock keeping households (including women) in the priority countries has increased their supply of livestock and livestock products to the market by through access to improved feed and forage. New business models developed for pellets/mash in Tunisia. Several cost-benefit analyses were conducted for forage technologies in Kenya and Colombia, indicating suitability of technologies is dependent on the market environment while strong contributions were made to innovation platforms in Colombia and Tunisia.
- **Technologies that reduce women livestock keepers' labour and energy expenditure:** Improved feeding practices reduce women's labour and energy expenditure by 10%. The improved practices disseminated reached about 1.6 million women in 12 countries.
- **Increased access to diverse nutrient-rich foods:** Poor people (men and women), in the priority countries benefited from increase in access to more affordable, safe and nutrient rich animal-source foods. About 30 million more people, of which 50% are women, meeting minimum dietary energy requirements (2019 Annual Report). In addition, 150 million more people, of which 50% are women, without deficiencies of one or more of the following essential micronutrients: iron, zinc, iodine, vitamin A, folate, and vitamin B12. National and international research and development partners and the private sector are using CRP developed forage and rangeland resources (with enhanced traits), in 30 countries and reaching producers who plant over 2 million ha, to increase the rate of genetic gain and exploit the genetic diversity of forages and rangeland species to enhance stress-tolerance, biomass productivity and nutritive value.

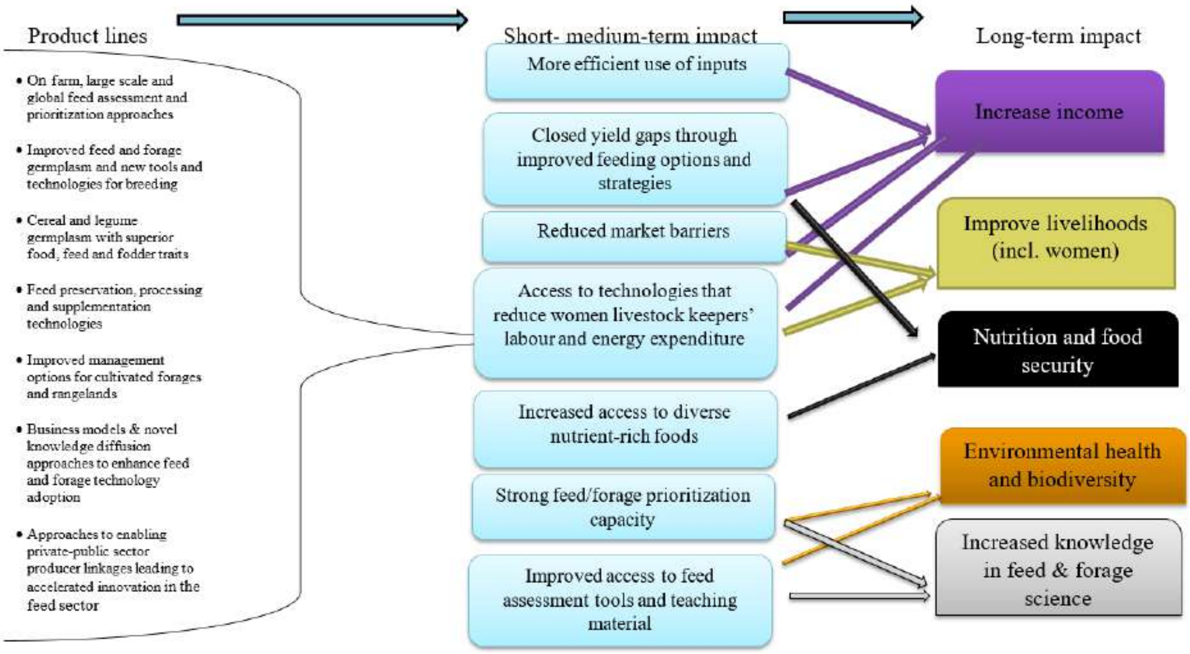


Figure 7. Detailed short-to-medium term and final impacts of Feed and Forage flagship

- **Improved access to feed assessment tools and teaching material:** This flagship has strengthen feed and forage prioritization. The flagship tested/applied the genderized Feed Assessment Tool (G-FEAST) in Burkina Faso, Uganda and Vietnam, launched the VegMeasure tool (for accurate measurement of rangelands/forest vegetation), Forage extension approaches were tested (Tunisia, Colombia) and training materials developed. For better utilization of feed/forage resources (3.5), feed processing options were adopted by public/private sector (India), and mobile seed cleaning/treatment units and fodder grinders were developed (Tunisia). Forage extension approaches were tested (Tunisia, Colombia) and training materials developed (2019 Annual Report). Extension approaches were tested (Kenya/Tanzania) and training materials developed in Haiti, Uganda, India, Kenya, Tanzania, and Colombia. A Colombian Sustainable Cattle Policy with contribution from the CRP is under revision by the government (2020 Annual Report). Access to data; at least 250,000 annual visitors to global databases, repositories, interactive tools and maps and the Tropical Grasslands/Forrajes Tropicales journal website. Furthermore, capacity building on CGIAR forage accessions for Latin American National Agricultural Research Systems (NARS) scientists, two international webinars on cactus pear organized by the Food and Agriculture Organization of the United Nations (FAO)-ICARDA network and virtual training on the CLEANED model for national partners in Ethiopia, Kenya, Nicaragua, Tanzania, Tunisia, Uganda and Vietnam was conducted.

In the long term, the impacts of the Feed and Forage flagship are the following:

- **Increased income:** More efficient use of inputs, closed yield gaps through improved feeding options and strategies and access to technologies that reduce women livestock keepers' labour and energy expenditure in the long run increases the income of livestock keeping households including women and youth involved in marketing of livestock products. The increase in income is also related to 20% increase in animal production using improved feed and forage technologies, a 10% accuracy increase for biomass and quality estimation (Annual Report 2019).

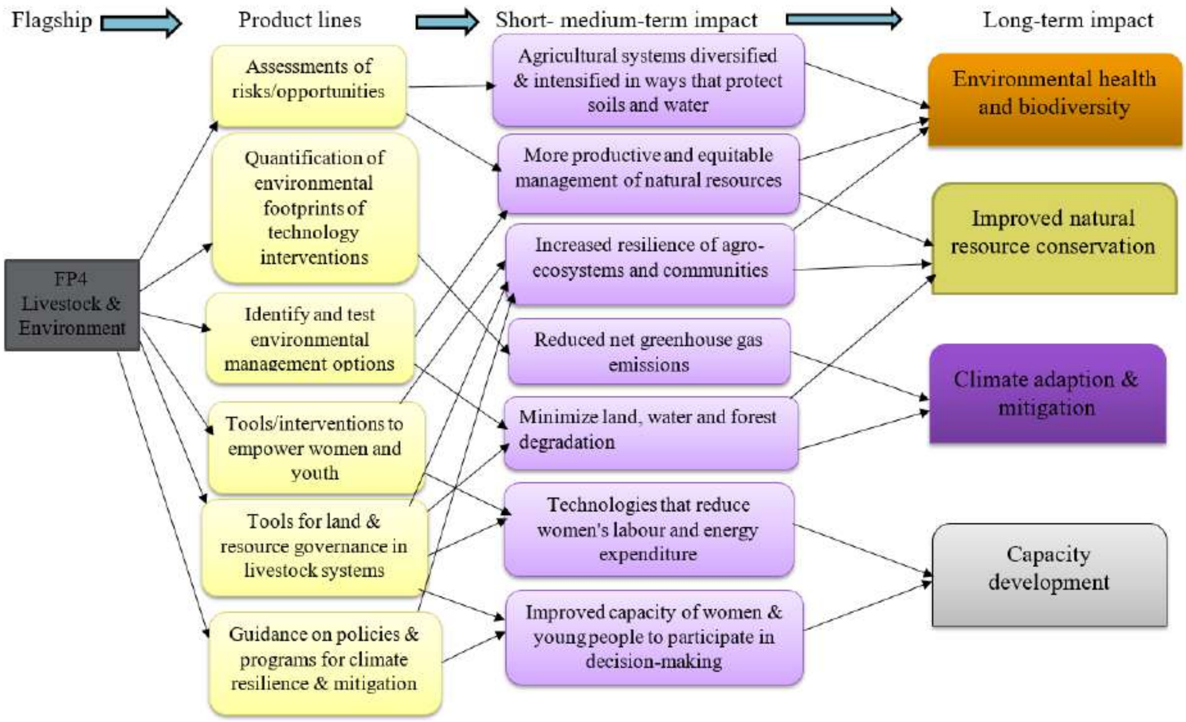
- **Improved livelihood:** Closing yield gaps through improved feeding options and strategies and access to technologies that reduce women livestock keepers' labour and energy expenditure improved the livelihood of livestock keeping households including women and youth. Livelihoods were also improved through increased delivery and uptake of feed and forage resources through proof-of-concept scaling, business model development and value-chain approaches by development partners, the private sector (feed and forage traders, feed processors) and (1 million by 2022) farmers across diverse environments in priority countries and other locations in Latin America, North and East Africa and South and Southeast Asia.
- **Nutrition and food security:** Closing yield gaps through improved feeding options and strategies and increased access to diverse nutrient-rich foods contributes to nutritional and food security in the priority countries. To improve livestock nutrition, the Feeds and Forages Flagship achieved further development and uptake of the suite of tools designed to make feed knowledge widely accessible and applied, while delivering a range of improved forages and dual-purpose crops, several to be distributed through private sector arrangements.
- **Environmental health and biodiversity:** Improved access to feed assessment tools and teaching material and improved access to feed assessment tools and teaching material contributed to environmental health and biodiversity. Feeds and Forages (FP3) produced a feasibility study on improved silvopastoral systems in the Caribbean region of Nicaragua, including ex-ante impact assessment of greenhouse gas emissions and carbon sequestration. Elsewhere, there was a four to five-fold variation in the amount of dry matter produced across the Napier grass population when grown under limited water supply with some accessions, which continued to grow well under severe water restrictions, potentially pushing the boundaries of where this forage crop can be grown (Annual Report 2019). The increase in biodiversity and environmental health is further related to increase land areas of forage cultivated. For example, a total of 100,000 new hectares planted in 2019 with forage hybrids in a total of at least 15 countries (calculated based on seed sales). The total area of hybrids scaled reached 1,000,000 hectares in 2019 and includes all hybrids scaled since 2001. CIAT's existing Urochloa hybrids were scaled on approximately 100,000 additional hectares in 15 countries in 2020. The total area sown with CIAT hybrids is estimated to be 1,100,000 hectares in 30 countries in 2020.
- **Increased knowledge in feed & forage science:** Improved access to feed assessment tools and teaching material and strong feed/forage prioritization capacity has led to increase in knowledge on feed and forage. The ICARDA Animal Feed Analysis Web Application (AFAWA) has reached at least 5,000 users by the end of 2018. Access of research partners to CRP generated knowledge on forages increased through 3 issues of the Tropical Grasslands journal (January, May, September) and 1 released and updated tool (SoFT). Feeds and Forages FP3 made significant advanced in 2018 on the development of feed intervention prioritization tools, which now incorporate cross-cutting issues such as gender. Regarding rangeland/pasture management [3.6], best-bet agronomic practices for cactus pear were developed/promoted (Asia/Africa). Feed Assessment Tool (FEAST) and Gendered (G)-FEAST) were applied in Vietnam, Kenya, Tanzania, Uganda. New Tropical Forages tool was launched, the AFAWA tool promoted, near-infrared spectroscopy (NIRS) equations/facilities developed, and the Sub-Saharan Africa feeds database updated. A 4-month weekly seminar series on sustainable cattle production, value chains and policies was also developed. While initially planned for a Colombian audience, the seminar series attracted an audience of 23,000 from 18 countries. Part of the seminar series dealt with the consequences and impacts of COVID-19 on the sector. Online trainings have so far reached over 120 participants who obtained ICARDA online certificates. The modules covered topics on cactus production, supplementary irrigation, innovation platforms, project development, bee keeping, and andragogy. Five two-

hour introductory workshops were organized for national extension and training staff to familiarize them with the new e-learning modules and show them how to register on the platform and navigate through the modules.

3.5.4. Livestock and Environment Flagship (FP4) Impact

Detailed short-to-intermediate term and final impacts attained from the Livestock and the Environment stream (FP4) are presented in Figure 8. Specifically, the short-to-intermediate term impacts of FP4 are the following:

- **Agricultural systems diversified and intensified in ways that protect soils and water:** The stream assessed risks/ opportunities arising from Global Environmental Change and developed materials to inform the global livestock and environment agenda in the priority countries. These activities contributed to diversification and intensification of livestock production systems in 50 locations in ways that protect soils and water, with benefits reaching 200,000 beneficiaries.
- **More productive and equitable management of natural resources:** Assessment of risks and opportunities arising from Global Environmental Change and developed materials to inform the global livestock and environment agenda as well as the identification and testing of environmental management options led to more productive and equitable management of natural resources. One hundred communities practice more productive and equitable management, with benefits experienced by 500,000 beneficiaries.
- **Reduced net greenhouse gas emissions from agriculture, forests and other forms of land use:** Quantification of environmental footprints of technology interventions and guidance on policies & programs for climate resilience and mitigation led to reduction in net greenhouse gas emissions from agriculture, forests and other forms of land use. Laboratory experiments with different feeding regimes to ascertain their effectiveness in reducing greenhouse gas intensities for mixed dairy and extensive pastoral systems (2019 Annual Report). Data



collection and publication of baselines on livestock production in Tanzania and Kenya. Incorporation of Tier 2 data from Kenyan systems into IPCC greenhouse gas emissions guidance/databases. Publication of a "best practice" brief summarizing experience to date by CCAFS and the Livestock CRP working with government partners to improve national reporting on greenhouse gas data for the livestock sector. Qualitative assessment of this stream and product lines indicated that GHG emission from agro-ecosystems in 5 priority locations is potentially reduced 10%, reaching 7 million beneficiaries (CCAFS).

Figure 8. Detailed short-to-medium term and final impacts of Livestock and Environment Flagship

- **Land, water and forest degradation (including deforestation) minimized and reversed:** The stream identified and tested environmental management options and developed tools for land and resource governance in livestock systems. These activities contributed to reduction in land and water degradation in 50 locations, positively impacting 200,000 beneficiaries. The flagship noted that government agencies in four countries are improving land governance arrangements to reduce degradation in rangelands.
- **Increased resilience of agro-ecosystems and communities, especially those including smallholders:** Resilience levels of agro-ecosystems and communities increased realized through the development of tools/interventions to empower women and youth, tools for land & resource governance in livestock systems and through guidance on policies and programs for climate resilience and mitigation. Agroecosystem resilience increased by 10% in 100 communities, affecting 500,000 beneficiaries.
- **Technologies that reduce women's labour and energy expenditure developed and disseminated:** Twenty environmental management interventions that reduce women's labour and energy expenditure were developed and disseminated. The development of tools/interventions to empower women and youth, tools for land & resource governance in livestock systems contributed to reduction in women's labour and energy expenditure.
- **Improved capacity of women and young people to participate in decision-making:** Tools for land and resource governance in livestock systems were developed in addition to guidance on policies and programs for climate resilience & mitigation. These tools, programs and policies incorporated gender and youth aspects in their development and hence improved the capacity of women and young people. This stream increased the capacity of 6,000 women and young people to participate in decision-making for environmental management of livestock.

In the long run, the impacts of FP4 can be summarized as follows:

- **Climate adaption & mitigation:** Tackling climate change was central to the mandate of Livestock and the Environment stream (FP4). This stream has contributed has contributed to climate adaption and mitigation through reduction in net greenhouse gas emissions and minimization of land, water and forest degradation. The flagship piloted feed interventions to reduce emissions in smallholder dairy systems across Tanzania, Rwanda, Kenya and Ethiopia. Advances were made for including soil organic carbon in greenhouse gas emissions assessments and feed interventions to reduce greenhouse gas emissions in smallholder dairy systems across Tanzania, Rwanda, Kenya and Ethiopia. Under the flagship, a key learning point was that East African dairy cattle show heat stress and decreased milk yields above temperature-humidity indexes of 67.5 (Annual report, 2019). This information was used to incorporate heat stress factors into the breeding objectives of dairy cattle breeding programs, allowing for future breeding of more heat-tolerant animals. The flagship also contributed to climate change adaptation by ensuring breeding programs produced genetically superior animals that were both productive and adaptive/resilient. The Livestock and the Environment Flagship, working with the Livestock Health Flagship, developed a new climate adaptation strategy by mapping heat stress for targeting tolerant breeds, and strengthened the evidence base on climate change risks and benefits associated with livestock in Africa.

- **Environmental health and biodiversity:** The diversification and intensification of agricultural systems that protect soils and water, more productive and equitable management of natural resources and increased resilience of agro-ecosystems and communities in priority countries contributed to improved and sustainable environmental health and biodiversity.
- **Improved natural resource conservation:** Ensuring more productive and equitable management of natural resources, minimization of land, water and forest degradation and increased resilience of agro-ecosystems and communities in priority countries contributed to improving the natural resource conservation in the priority countries.
- **Capacity development:** Livestock and the Environment stream contributed to capacity development of women and young people in the priority countries. This was realized through the use of development and use of ttechnologies that reduce women's labour and energy expenditure as well as improvement in the capacity of women and young people to participate in decision-making. The flagship issued guidance for researchers on engaging policy makers in improving greenhouse gas emissions data in smallholder dairy systems across Tanzania, Rwanda, Kenya and Ethiopia, and conducted trainings on the Rural Household Multi Indicator Survey (RHoMIS) tool, which enables evaluation of trade-offs between household food security, production and environmental objectives.

3.5.5. Livestock Livelihoods and Agri-Food Systems Flagship (FP5) Impact

Detailed short-to-intermediate term and final impacts realized from Livestock Livelihoods and Agri-Food Systems stream are outlined in Figure 9. The short-term and intermediate impacts are the following:

- **Gender-equitable control of productive assets and resources:** Activities under gender and social equity productive line led to gender-equitable control of productive assets and resources. Adequately ensuring the participation of women in this stream helps to bridge the gaps hindering their access to assets and services. For example, expanding forage production by women is often constrained by their long-term ability to access land, so they need to be included in developing solutions. Women’s Empowerment in Livestock Index (methodology and application) was developed and adopted as a monitoring tool. The use of the Women's Empowerment in Livestock Index (WELI) resulted in a larger WELI database and the ability to compare data across 6 countries (Nepal, Senegal, Uganda, Ethiopia, Kenya, Ghana and Tanzania). Field research on gender transformative approaches commenced. Gender equity relative to their level of effort (i.e. labour) at household level in the use of, and control of income generated by, livestock related productive assets and resources, impacting 288,000 women across 4 countries. WELI measures women’s empowerment in agriculture, focusing on key areas of livestock production (e.g. animal health, breeding and feeding) and on use of livestock products (e.g. animal-source-food processing and marketing).
- **Improved capacity of women and youth to participate in decision-making:** Effectively ensuring the involvement of women and youth helps to bridge the gaps hindering their participation in decision making, services, and information that are key to technology adoption. Increasing gender investments in livestock at a national level requires a new approach to gender that integrates women into the planning, as most investments are based on herd models. This was a clear finding of recent work in Bihar, India where gender was integrated into the Livestock Master Plan. This stream potentially improved capacity of one million women and young people to participate in livestock related decision-making in 5

countries. The Women's Empowerment in Livestock Index, created by ILRI and Emory University, was adopted by four projects under the Livestock Vaccine Innovation Fund of the International Development Research Centre, providing harmonized data to measure the empowerment of women as users, entrepreneurs and service providers in the livestock vaccine distribution/delivery chain. The tool is being used by a major international funder at scale, reaching women participants across six countries (Kenya, Uganda, Rwanda, Senegal, Ghana, and Nepal). Elevating gender considerations into policies and investment decisions is done through the livestock master plans and associated modelling.

- **Increased access to diverse nutrient-rich foods:** Food and nutrition security through livestock product lines and activities contributed to increase access to diverse nutrient-rich foods. About 3 million poor people (men and women), in 4 countries, with increase in access to more affordable, safe and nutrient rich animal-source foods.
- **Increased household capacity to cope with shocks:** Integrated technologies, practices and institutions for improved livestock systems and food and nutrition security through livestock contributed to increased household capacity to cope with shocks. Innovative institutional options that improve resilience tested and adopted by national and international research & development partners, increasing the resilience of 356,000 rural livestock-keeping households (1.7 million individuals) in 3 countries.
- **Increased livelihood opportunities:** Integrated technologies, practices and institutions for improved livestock systems created more livelihood opportunities. Livelihood opportunities increased about 15%, on average in total household income from livestock-related activities, including 25% increase, on average, in proportion controlled by women, for 449,000 households in 8 countries. In 2019, EAT Lancet Report introduced new global dietary guidelines that promoted reduced consumption of animal source foods. In response, scientists from the CGIAR research programs on Livestock and Agriculture for Nutrition and Health (A4NH) synthesized and promoted evidence on the contribution of livestock to human nutrition and livelihoods in developing countries. Livestock Livelihoods and Agri-Food Systems FP5 partner organization, Heifer International reported the number of farmers who are using dairy hubs as 8,717 in Tanzania, 17,824 in Uganda and 60,081 in Kenya (2018 Annual Report).
- **Reduced market barriers:** Integrated technologies, practices and institutions for improved livestock systems and policies, foresight & systems analysis contributed to reducing market barriers in the priority countries. About 454,000 livestock keeping households (representing 2.2 million individuals, including women) increase their supply of livestock to the market by 15%, on average, in 7 countries.

- **Conducive agricultural policy environment:** Policies, foresight and systems analysis contributed to laws, rules and regulations within and across 4 countries at local, country and regional level explicitly included pro-poor livestock mediated development, reaching 2 million livestock keepers & other value-chain actors.

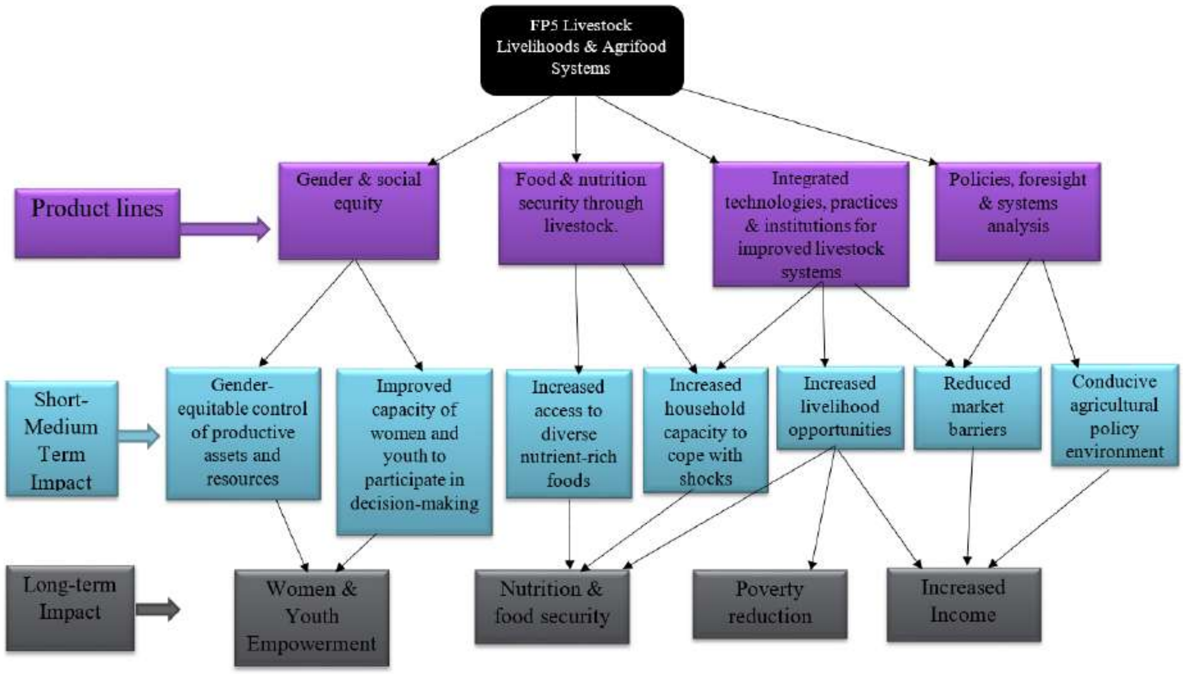


Figure 9. Detailed short-to-medium term and final impacts of the Livestock Livelihoods & Agri-Food Systems flagship

In the long-term, the following are the main impacts of the Livestock Livelihoods & Agri-Food Systems flagship

- **Women and youth empowerment:** Gender-equitable control of productive assets and resources and improvement in capacity of women and youth to participate in decision-making contributed to women and youth empowerment in the priority countries.
- **Nutrition & food security:** Increased access to diverse nutrient-rich foods, household capacity to cope with shocks and livelihood opportunities contributed to improvement in nutritional and food security in the priority countries. IILRI and its partners have implemented a social and behaviour change communication intervention on animal source foods consumption, particularly the drinking of milk among Girinka households in Rwanda. This was implemented through a Feed the Future Innovations Labs for Livestock Systems (LSIL) research project. In 2020 the national counselling cards were updated to include animal source foods consumption messages, which have been validated through the National Child Development Agency and are now being used nationally by community health workers.
- **Poverty reduction:** Increased livelihood opportunities and reduction in market barriers through this stream contributed to reduction in poverty in the priority countries.
- **Increased Income:** Increased livelihood opportunities, reduced market barriers and the creation of conducive agricultural policy environment under the Livestock Livelihoods and Agri-Food Systems stream contributes to increase in income.

3.6. Sustainability

The CRP Livestock incorporated strategies to ensure sustainability of streams/flagships' impacts at the end of the project. To begin with, the CRP livestock established collaboration with national and international development partners, government agencies and extension systems, including technology developers. This collaboration ensures the continuation of the streams activities and its impacts beyond the end of the CRP livestock program period. Further advancements of the CRP can be maintained through government agencies and development partners, at local and national levels across priority countries and other locations. These partners are promoting the CRP product lines and activities across different priority countries. Another sustainability strategy is the incorporation of CRP livestock streams' technologies, tools and concepts into teaching materials and curriculum of universities and research institutions in the priority countries. Students and researchers are the future farm advisors, policymakers and industry representatives, making it important to communicate and transfer the knowledge to future generation. Progress was also made towards the milestone of having technology developers taking environmental issues into account in research priority setting.

4. Conclusions and considerations for future VfM assessments of CRP Livestock

Generally, each of the flagships was assessed for short-medium term and final impacts. Detailed and different TOCs were created to show the pathways through which each flagship's final impacts were attained. Although we did not quantify the final impacts realized from each flagship, it is important to conclude that the economic, food and nutritional security, poverty alleviation, environmental, climate change adaptation, capacity building, education, and inclusiveness dimensions differ across research streams. Specifically, the long term impacts of the Animal Genetics flagship (FP1) include (a) contribution to poverty reduction, (b) contribution to hunger and food insecurity reduction, (c) income, (d) education, knowledge and innovation, (e) capacity development. The Livestock Health (FP2) flagship in the long term contributed to (a) improved health of both human and animals (b) food/nutritional security (c) increase income of poor livestock keepers including women and youth (d) improved livelihoods livestock keeping households and communities (e) education/capacity development. The long term impacts of Feeds and Forages (FP3) include (a) increase in income of poor livestock keepers (b) improved livelihood of livestock keeping households and communities (c) enhanced nutritional and food security (d) improved environmental health and biodiversity (e) increased knowledge in feed & forage science. In the long term, the Livestock and Environment flagship (FP4) contributed to (a) climate adaption and mitigation (b) improved environmental health and biodiversity (c) improved natural resource conservation (d) capacity development. The Livestock Livelihoods & Agri-Food Systems (FP5) in the long term contributed to (a) Women and youth empowerment (b) Nutrition & food security (c) Poverty reduction (d) Increased Income. The short timeframe and unavailability of evaluation data has not allowed a fully harmonized estimation of causality parameter estimates in terms of magnitude of impacts attributed to each research stream. To quantify the impact of the CRP livestock program using the scoring methods and value for money approach will require additional qualitative and quantitative data aggregated across streams.

With regard to considerations for future VfM assessments of CRP Livestock, it is important to highlight that impact evaluation of the CRP Livestock, like any evaluation, will largely be most reliable and valid when it uses a mixed methods approach where both qualitative and quantitative data are used to establish evidence for impact. Typically data will be assembled that supports (or refutes) a claim about the extent to which a programme was responsible for delivering an intended outcome. The main limitation of the current CRP livestock impact assessment nature of the CRP does not readily allow quantification, i.e. measurement of the attributable impact of an intervention. In this assignment, we could not quantify the actual or precise impacts from different CRP streams because of data limitations. CRP currently does not have data on number of beneficiaries of each research stream and product lines and as such it is difficult to quantify the magnitude of impacts from the CRP streams. Therefore, we produce data considerations and requirements for assessing CRP livestock impact in the future. In Table 2, qualitative and quantitative data required needed for a more rigorous or preferred value-for-money methodology that would cover the full scope of upstream-to-downstream research activities.

The lack of end-line data is due to several factors. To begin with, the broad nature of the CRP makes it difficult to quantify the actual impact and the number of beneficiaries from each product line activity and priority countries. Social and political factors in some priority countries affect data gathering. For instance, Feeds and Forages Flagship activities in Nicaragua were severely affected by the socio-political unrest in April 2018 and a planned cost-benefit analysis had to be dropped (Annual Report, 2018). The pandemic's associated travel and work restrictions delayed many field and laboratory activities, requiring research teams to adjust their research designs and schedules and to rely more heavily on local partners to implement field work. The most adversely affected was the CRP's cross-flagship experiment in translating research results into design and piloting livestock development interventions in its four priority countries, which was already under a tight schedule given its stalled start and shortened horizon. Delays imposed by the pandemic have limited testing and the evidence that can be generated in the remaining time. This affected data gathering data for the last two years of the program.

Table 2. Data requirements for value-for-money methodology

Research stream/Flagship	Final Impacts	Qualitative data required	Quantitative data required
Animal Genetics (FP1)	i. Poverty Reduction ii. Hunger/food insecurity reduction iii. Income iv. Education/innovation v. Capacity pment	i. Requires qualitative experts review data on scoring/ranking of each outcome indicators highlighted in the FP1 TOC. This evaluation and scoring can be done by the stream leader(s), research leaders of product lines and activities as well as sampled beneficiaries. Scoring methods can be used where measuring scales are used to allocate weights, ranks and benchmarks to the streams' identified impacts. The scoring data can be used to generate composite index consisting of all weighted indexes for the identified final impacts. ii. Participatory methods can be used to allow groups to identify changes resulting from the project, who has	i. Requires baseline and follow-up survey data collected on impact indicators across the stream's product lines /activities ii. Requires quantitative data on the number of beneficiaries (i.e. individual, households, communities, etc.) iii. Requires quantitative data on the proportion of change recorded due to the stream (e.g. % reduction in poverty, food security, etc.) iv. Requires parallel and sequential quantitative data gathering

Research stream/Flagship	Final Impacts	Qualitative data required	Quantitative data required
		benefited and by what proportion. iii. Triangulation can be used to compare the group information with the opinions of key informants and information available from secondary sources. iv. Case studies on individuals or groups may be produced to provide more in-depth understanding of the processes of change	on the impacts outlined in the TOC. The quantitative data on each impact indicators should be documented together with the qualitative data from all activities across the stream.
Animal Health (FP2)	i. Improved health ii. Nutritional/food security iii. Increase income iv. Improved livelihood v. Education/capacity development	i. Requires qualitative experts review data on scoring/ranking of each outcome indicators highlighted in the FP1 TOC. This evaluation and scoring can be done by the stream leader(s), research leaders of product lines and activities as well as sampled beneficiaries. Scoring methods can be used where measuring scales are used to allocate weights, ranks and benchmarks to the streams' identified impacts. The scoring data can be used to generate composite index consisting of all weighted indexes for the identified final impacts. ii. Participatory methods can be used to allow groups to identify changes resulting from the project, who has	i. Requires baseline and follow-up survey data collected on impact indicators across the stream's product lines /activities ii. Requires quantitative data on the number of beneficiaries (i.e. individual, households, communities, etc.) iii. Requires quantitative data on the proportion of change recorded due to the stream (e.g. % reduction in poverty, income, food security, etc.) iv. Requires parallel and sequential quantitative data gathering on the impacts outlined in the TOC. The

Research stream/Flagship	Final Impacts	Qualitative data required	Quantitative data required
		benefited and by what proportion. iii. Triangulation can be used to compare the group information with the opinions of key informants and information available from secondary sources. iv. Case studies on individuals or groups may be produced to provide more in-depth understanding of the processes of change	quantitative data on each impact indicators should be documented together with the qualitative data from all activities across the stream.
Feeds and Forages Flagship (FP3)	i. Increase income ii. Improved livelihood iii. Nutrition and food security iv. Environmental health and biodiversity: v. Increased knowledge in feed & forage science	i. Requires qualitative experts review data on scoring/ranking of each outcome indicators highlighted in the FP1 TOC. This evaluation and scoring can be done by the stream leader(s), research leaders of product lines and activities as well as sampled beneficiaries. Scoring methods can be used where measuring scales are used to allocate weights, ranks and benchmarks to the streams' identified impacts. The scoring data can be used to generate composite index consisting of all weighted indexes for the identified final impacts. ii. Participatory methods can be used to allow groups to identify changes resulting from the project, who has	i. Requires baseline and follow-up survey data collected on impact indicators across the stream's product lines /activities ii. Requires quantitative data on the number of beneficiaries (i.e. individual, households, communities, etc.) iii. Requires quantitative data on the number of adopters (i.e. livestock keeper, households, communities, etc.) iv. Requires quantitative data on the proportion of change recorded due to the stream (e.g. % reduction in poverty, income,

Research stream/Flagship	Final Impacts	Qualitative data required	Quantitative data required
		benefited and by what proportion. iii. Triangulation can be used to compare the group information with the opinions of key informants and information available from secondary sources. iv. Case studies on individuals or groups may be produced to provide more in-depth understanding of the processes of change	food security, etc.) v. Requires parallel and sequential quantitative data gathering on the impacts outlined in the TOC. The quantitative data on each impact indicators should be documented together with the qualitative data from all activities across the stream.
Livestock and Environment Flagship (FP4)	i. Climate adaption & mitigation v. Environmental health and biodiversity vi. Improved natural resource conservation vii. Capacity development	i. Requires qualitative experts review data on scoring/ranking of each outcome indicators highlighted in the FP1 TOC. This evaluation and scoring can be done by the stream leader(s), research leaders of product lines and activities as well as sampled beneficiaries. Scoring methods can be used where measuring scales are used to allocate weights, ranks and benchmarks to the streams' identified impacts. The scoring data can be used to generate composite index consisting of all weighted indexes for the identified final impacts. ii. Participatory methods can be used to allow	i. Requires baseline and follow-up survey data collected on impact indicators across the stream's product lines /activities ii. Requires quantitative data on the number of beneficiaries (i.e. individual, households, communities, etc.) iii. Requires quantitative data on the number of adopters (i.e. livestock keeper, households, communities, etc.) iv. Requires quantitative data on the proportion of change recorded due to the stream (e.g. %

Research stream/Flagship	Final Impacts	Qualitative data required	Quantitative data required
		groups to identify changes resulting from the project, who has benefited and by what proportion. iii. Triangulation can be used to compare the group information with the opinions of key informants and information available from secondary sources. iv. Case studies on individuals or groups may be produced to provide more in-depth understanding of the processes of change	reduction in poverty, income, food security, etc.) v. Requires parallel and sequential quantitative data gathering on the impacts outlined in the TOC. The quantitative data on each impact indicators should be documented together with the qualitative data from all activities across the stream.
Livestock Livelihoods and Agri-Food Systems Flagship (FP5)	i. Women and youth empowerment vi. Nutrition & food security vii. Poverty reduction viii. Increased Income	i. Requires qualitative experts review data on scoring/ranking of each outcome indicators highlighted in the FP1 TOC. This evaluation and scoring can be done by the stream leader(s), research leaders of product lines and activities as well as sampled beneficiaries. Scoring methods can be used where measuring scales are used to allocate weights, ranks and benchmarks to the streams' identified impacts. The scoring data can be used to generate composite	i. Requires baseline and follow-up survey data collected on impact indicators across the stream's product lines /activities ii. Requires quantitative data on the number of beneficiaries (i.e. including women, youth, children, etc.) iii. Requires quantitative data on the number of adopters (i.e. livestock keeper, households,

Research stream/Flagship	Final Impacts	Qualitative data required	Quantitative data required
		index consisting of all weighted indexes for the identified final impacts. ii. Participatory methods can be used to allow groups to identify changes resulting from the project, who has benefited and by what proportion. iii. Triangulation can be used to compare the group information with the opinions of key informants and information available from secondary sources. iv. Case studies on individuals or groups may be produced to provide more in-depth understanding of the processes of change	communities, etc.) iv. Requires quantitative data on the proportion of change recorded due to the stream (e.g. % reduction in poverty, income, food security, etc. v. Requires parallel and sequential quantitative data gathering on the impacts outlined in the TOC. The quantitative data on each impact indicators should be documented together with the qualitative data from all activities across the stream.

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Appendix

Table A. CRP Livestock flagships and product lines

Flagships	Product Lines
Flagship #1: Animal Genetics (4 product lines)	1.1 Genomic selection based breeding programs for cross-bred dairy cattle, linked to natural sire and artificial insemination based delivery systems, in Tanzania & Ethiopia
	1.2 Community based breeding programs for small ruminants, linked to natural sire and artificial insemination based delivery systems, in Tanzania and Ethiopia.
	1.3 Genetic improvement programs based on breed substitution for chickens, linked to distribution systems for the improved breeds in Tanzania, Ethiopia & Nigeria
	1.4 Genetic improvement strategy for pigs, linked to community based artificial insemination delivery systems, in Uganda.
Flagship #2: Animal Health (7 product lines)	2.1 Assessment tools for significance of animal diseases
	2.2 Risk models and maps for animal diseases
	2.3 Herd health packages
	2.4 Medically rational use of antimicrobials
	2.5 Novel vaccines
	2.6 Novel diagnostics
	2.7 Improved access to animal health services and products
Flagship #3: Feeds and Forages (7 product lines)	3.1 On farm, large scale and global feed assessment and prioritization approaches
	3.2 Improved feed and forage germplasm and new tools & technologies for breeding
	3.3 Cereal and legume germplasm with superior food, feed and fodder traits
	3.4 Feed preservation, processing and supplementation technologies
	3.5 Improved management options for planted/cultivated forages and rangelands
	3.6 Business models and novel knowledge diffusion approaches to enhance feed and forage technology adoption
	3.7 Approaches to enabling private-public sector producer linkages leading to accelerated innovation in the feed sector
Flagship #4: Environment (6 product lines)	4.1 Assessments of risks/ opportunities arising from Global Environmental Change and materials to inform the global livestock and environment agenda
	4.2 Quantification environmental footprints of technology interventions
	4.3 Identify and test environmental management options; quantify benefits.
	4.4 Tools/ interventions to empower women and youth as agents of change for environmental management
	4.5 Tools for land and resource governance in livestock systems
	4.6 Guidance on policies and programs for climate resilience and mitigation
Flagship #5: Livestock Livelihoods & Agri-food Systems (1 product line)	5.1 Macro level analyses of livestock markets and systems at global and regional scale

Table B. Example of evaluation questions and data sources for Flagship #1

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
RELEVANCE			
KEQ1: How well was FP1 programme design suited to its objectives, the context, and the needs of its target group?	DEQ1.1: Is the FP1 ToC valid and comprehensive relative to what was required for FP1?	▪ The FP1's ToC articulates well-sequenced, coherent and clear causal pathways and makes explicit the most critical assumptions ▪ The underlying assumptions in the ToC are plausible based on available evidence ▪ FP1's product lines are consistent with the programme's intended outcomes and impact ▪ FP1 product lines are consistent with good practice, and build on previous experience of what has worked and not ▪ Key common bottlenecks and barriers are addressed as part of this flagship activities	
	DEQ1.2: How relevant was the FP1 to the local sectors and contexts?	▪ The FP1 aligns with national and provincial government priorities and policies ▪ The FP1 understands local systems and adapts to these systems as needed ▪ This flagship's objectives and interventions are supported by key actors and government agencies	
	DEQ1.3: How relevant was the FP1 to the local research/academic institutions?	▪ The FP1 aligns with local research and educational institutions priorities and policies	

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
		▪ The FP1 understands local research and educational systems and adapts to these systems as needed ▪ This flagship's objectives and product lines were supported by key research and academic institutions	
	DEQ1.4: How relevant was the FP1 to the needs of the target population subgroups?	▪ Latest available evidence supports potential for impact of the FP1's proposed approach ▪ The FP1 aligns with policy priorities related to addressing the special needs of target groups	
	DEQ 1.5: How successfully has the FP1 adapted to the context of implementation and newly available evidence?	▪ FP1 adequately monitors programme implementation, results and the market & policy context, and makes information accessible ▪ The programme takes into account new evidence on effectiveness of the FP1 flagship activities and implements programme adjustments accordingly ▪ The programme's ToC was periodically reviewed and adjusted when needed to emerging evidence about how the theory holds in practice	
COVERAGE			
KEQ2: How well did FP1 reach its target population subgroups?	DEQ2.1: How many poor livestock keepers and households benefited from FP1?	▪ The number of poor livestock households that benefited from the four product lines under this FP1.	

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
		▪ The number of poor livestock farmers that benefited from the four product lines under this FP1.	
	DEQ2.2 How many women and men benefited from FP1?	▪ The number of men and women reached by the four product lines under this FP1.	
	DEQ2.3 How many youth were reached by FP1?	▪ FP1 adequately incorporated the needs of youth people in the study areas	
	DEQ2.4: To what extent do poor and other vulnerable groups consume livestock products from the newly developed livestock breeds and poultry improvement?		
EFFECTIVNESS			
KEQ3: To what extent has FP1 contributed to an adequate supply of livestock and poultry products?	DEQ3.1: To what extent is the newly improved animal genomic and genetic resources are used by the small-medium and large scale producers targeted by the programme?	The programme's ToC takes into the need for capacity building.	
	DEQ3.2: How many curricula and training modules were developed?		
	DEQ3.3: How many innovative learning materials and approaches were developed?		
	DEQ3.4. How many students, researchers (e.g. BSc, Msc, PhD) benefited from each product line?	The programme's ToC takes into account the need for capacity building.	
	DEQ3.5: What other factors influence the		

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
	production and distribution of improved animal genomic and genetic resources?		
	DEQ3.6: To what extent is a sustainable supply of improved animal genomic and genetic resources available in markets/breeding stations/retail outlets?		
	DEQ3.4: What factors influence the sustainable supply of improved animal genomic and genetic resources in markets		
KEQ4: To what extent has FP1 contributed to raising public awareness and acceptance of improved animal genomic and genetic resources, and its benefits?	DEQ4.1: To what extent has the FP1 public awareness activities contributed to raising awareness of improved animal genomic and genetic resources, and its benefits?		
	DEQ4.2: To what extent has the FP1 public awareness activities contributed to more acceptance and consumption of livestock and poultry products?		
	DEQ4.3: What other factors influence consumers' awareness and acceptance of, and willingness to purchase, newly improved livestock and poultry products?		
KEQ5: To what extent has, FP1 contributed to an improvement in public sector management of the	DEQ5.1 To what extent has the FP1 contributed to making use and conservation of animal genomic and		

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
animal genomic and genetic resources in accordance with mandatory legislation and revised standards and regulations?	genetic resources mandatory, and to the adoption of revised and harmonised regulations and standards?		
	DEQ5.2 To what extent has the FP1 contributed to the government improving monitoring and enforcement of animal genomic and genetic resources regulations and standards?		
	DEQ5.3 To what extent has the FP1 contributed to building awareness of, and political commitment and support for animal genomic and genetic resources development and adoption?		
	DEQ5.4 What other factors influence political commitment, support, and improved public sector management of animal genomic and genetic resources development?		
EFFICIENCY			
KEQ6: Is FP1 cost effective and does it offer VfM?	DEQ6.1: Are the new genetic resources and breeds cost-effective compared to business-as-usual use of traditional genetic resources and breeds in the study areas?		
	DEQ6.2: What proportion of income gains was realized in this product line?		

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
	DEQ6.3: For each of the product lines, how many new breeds were introduced?		
	DEQ6.4: What are the prolificacy of these new breeds? List for each breed.		
	DEQ6.5: What are the percentage improvement in animal weight gained for the new breeds?		
	DEQ6.6: How many community based breeding programs were implemented under each product line		
	DEQ6.7: What is the yield/output potential of each new breed listed above		
	DEQ6.8: What efficiency gains were achieved under each product line?		
ECONOMY/IMPACT			
KEQ7: To what extent has FP1 improved the consumption of adequately newly improved livestock and poultry products and estimated nutritional status of the target population?	DEQ7.1: To what extent has the micronutrient (e.g. protein) intake of rural poor households increased due to the consumption of newly improved livestock and poultry products?	The programme takes into account nutritional needs of the poor, particularly in rural area.	
	DEQ7.2: What are the predicted improvements in the micronutrient status of children in different study provinces due to the consumption of newly improved livestock and poultry products produced?		
	DEQ7.3. In terms of poverty reduction, by		

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
	what proportion has the different product line reduced poverty in the study areas?		
	DEQ7.4: What are the key factors that facilitate or inhibit the consumption of newly improved livestock and poultry products, particularly among the poor and children under; and how do consumers experience these factors?		
KEQ8: How has FP1 influenced the market system of improved livestock and poultry products?	DEQ8.1 To what extent and how has the introduction of newly improved livestock and poultry genetic resources and breeds affected business performance and practices in the value chain?		
	DEQ8.2: What effect has the programme had on the prices and perceived affordability of newly improved livestock and poultry products?		
	DEQ8.3: To what extent did the programme influence the newly improved livestock and poultry products market?		
SUSTAINABILITY			
KEQ9: To what extent is it likely that FP1 will lead to a continuation of large-scale production of livestock and poultry products after the programme ends?	DEQ9.1: What factors are likely to affect the continuation of large-scale production of improved livestock and poultry products using genomic and genetic resources after the programme ends?		

Key evaluation questions (KEQs)	Detailed evaluation questions (DEQs)	Criteria to answer the questions	Data sources/files
	DEQ9.2: To what extent are factors that are likely to support or inhibit the sustainability of large-scale production of livestock and poultry products put in place or addressed?		